SAIR

Spatial AI & Robotics Lab

# CSE 473/573

## L17: FACE DETECTION

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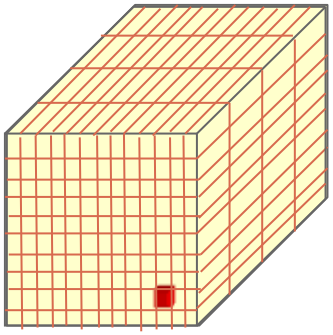
# Why is detection hard?



We want to do this for ~ 1000 objects

1,000,000 images/day

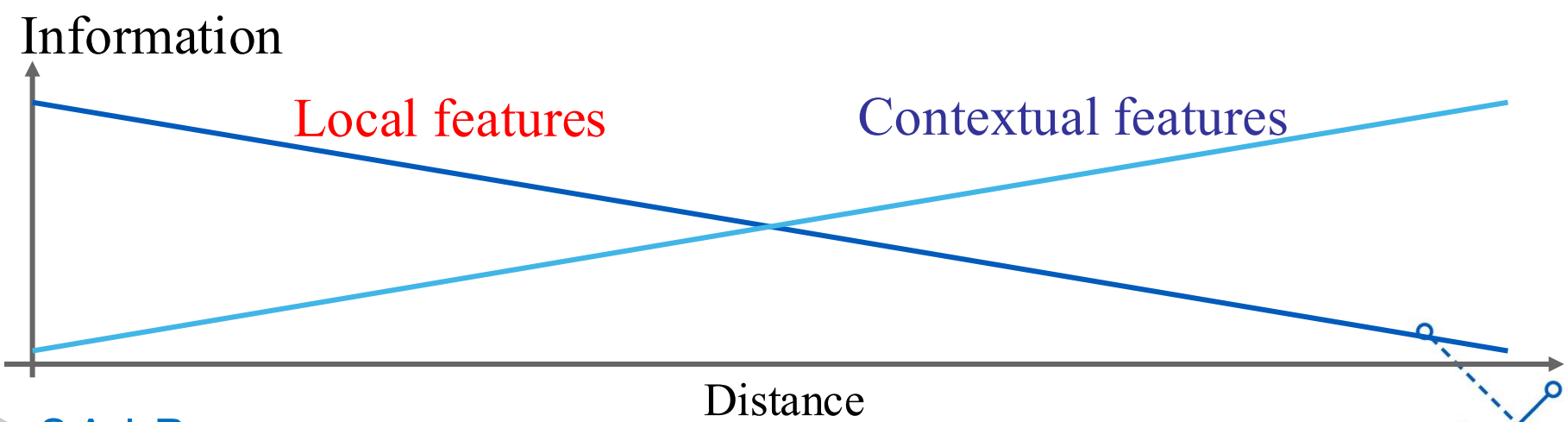
# Why is detection hard?

If we have 1000 categories (detectors), and each detector produces 1 false every 10 images, we will have 100 false alarms per image… pretty much garbage…

# Local information is helpful

Slide credit: A. Torralba

# Is local information enough?






Slide credit: A. Torralba

# Is local information even enough?

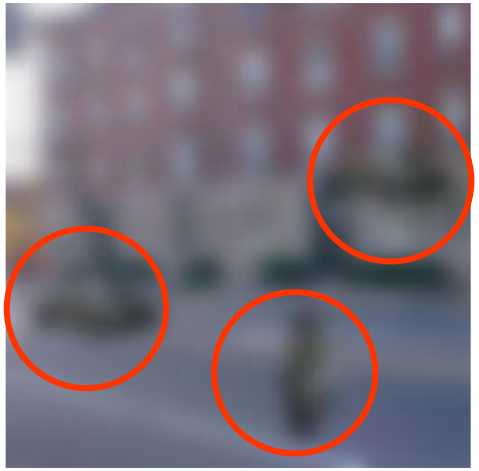
We know there is a keyboard present in this scene even if we cannot see it clearly.

We know there is no keyboard present in this scene

**… even if there is one indeed.**

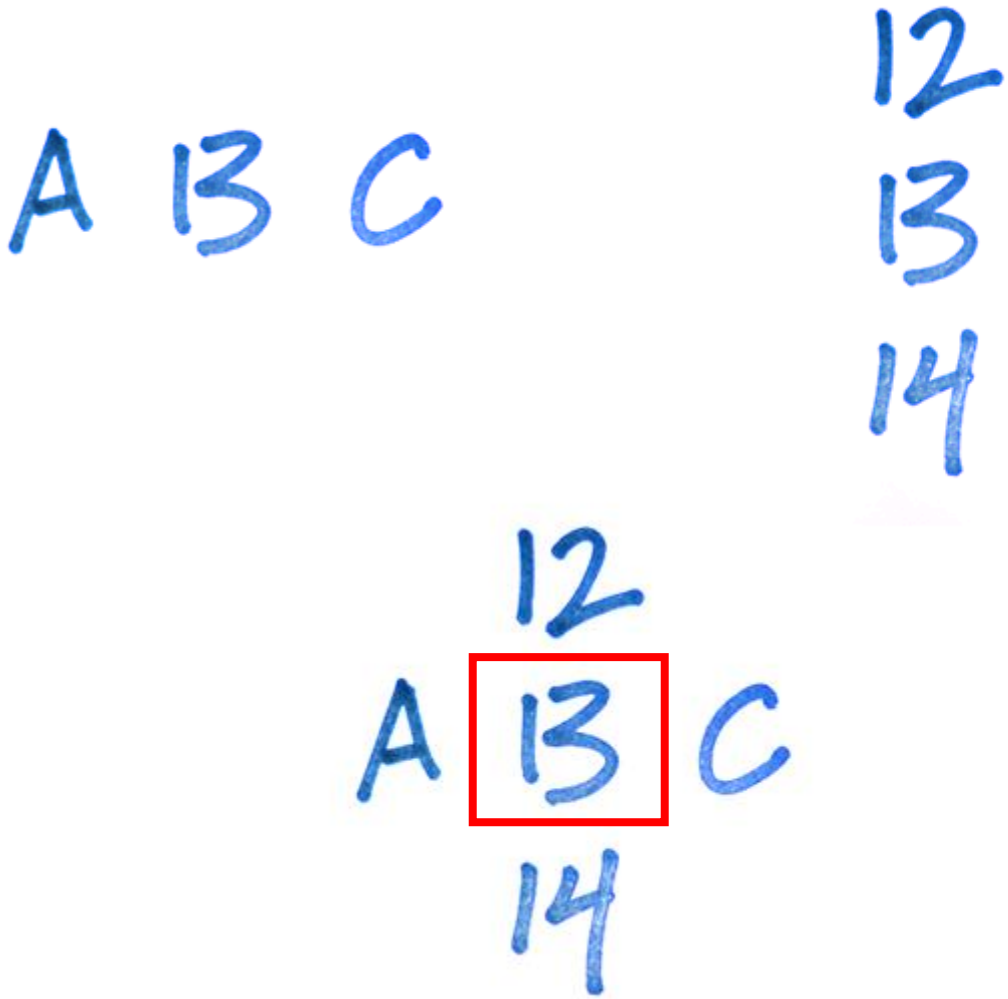
Slide credit: A. Torralba

# The multiple personalities of a blob



Slide credit: A. Torralba

# The multiple personalities of a blob



# The context challenge

- What are the hidden objects?

Slide credit: A. Torralba

# The context challenge

- What are the hidden objects?

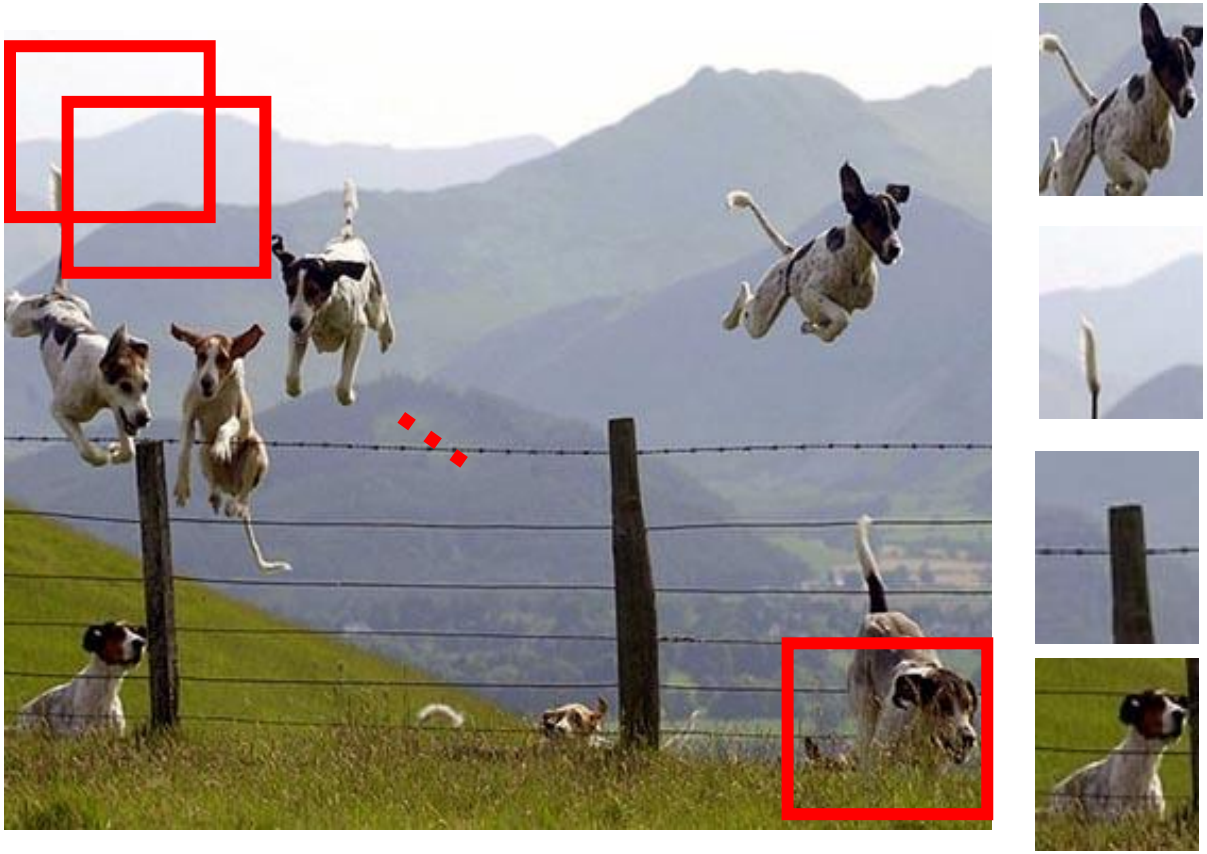
Slide credit: A. Torralba

# Object detection vs Scene Recognition

- Objects (even if deformable and articulated) probably have more **consistent shapes** than scenes.
- Scenes can be defined by distribution of “stuff” – materials and surfaces with **arbitrary shape**.
- Objects are **“things” that own their boundaries**
- Bag of words (BoW) were less popular for object detection because they **throw away shape** info.

# Object Category Detection

- Focus on object search: “Where is it?”
- Build templates that quickly differentiate object patch from background patch



Dog Model

**Object** or **Non-Object?**

# Challenges in modeling the object class

Illumination

Object pose

Clutter

Occlusions

Intra-class appearance

Viewpoint

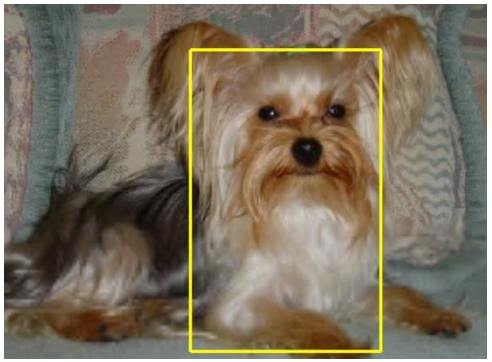


Slide from K. Grauman, B. Leibe

# Challenges in modeling the non-object class

True Detections

Bad Localization



Confused with Similar Object

Misc. Background

Confused with Dissimilar Objects

# General Process of Object Recognition

Specify Object Model

What are the object parameters?

↓

Generate Hypotheses

↓

Score Hypotheses

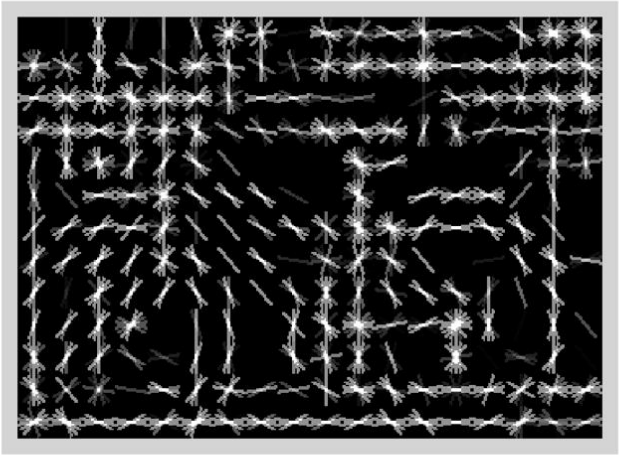
↓

Resolve Detections

# Specifying an object model

1. Statistical Template in Bounding Box
   - Object is somewhere (x,y,w,h) in image
   - Features defined wrt bounding box coordinates

Image

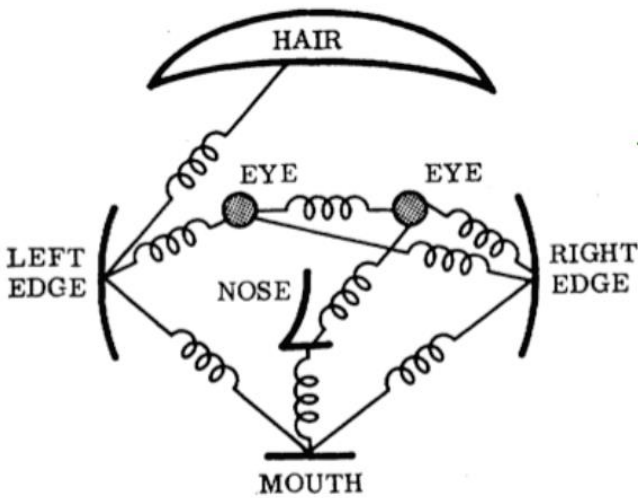


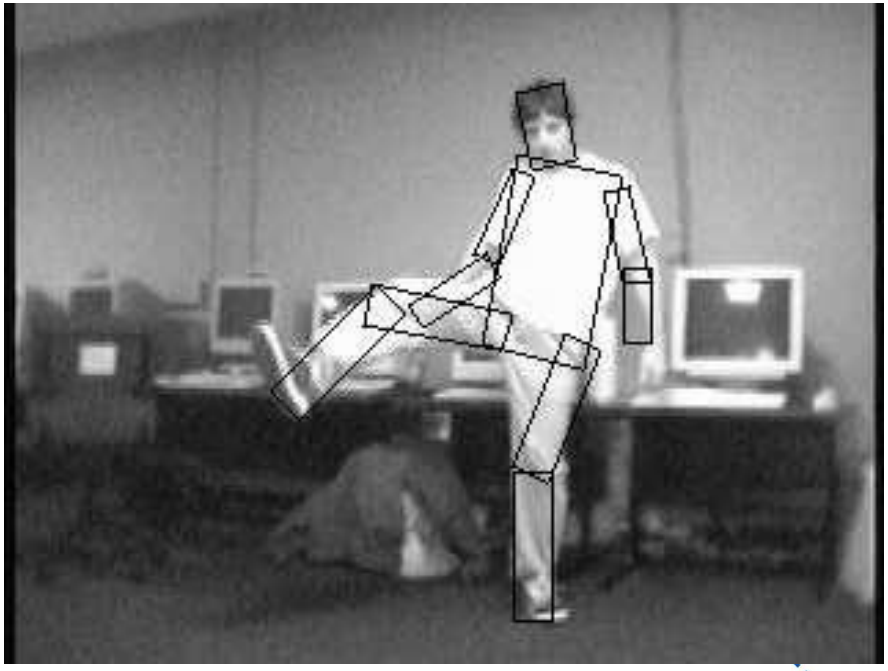
Template Visualization
(HoG features)

Images from Felzenszwalb

# Specifying an object model

2. Articulated parts model
   - Object is configuration of parts
   - Each part is detectable



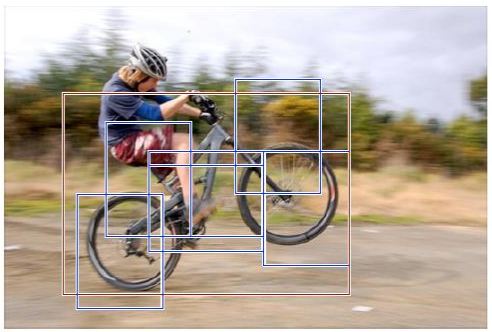



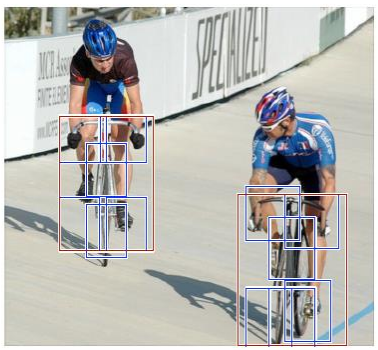
Images from Felzenszwalb

# Specifying an object model

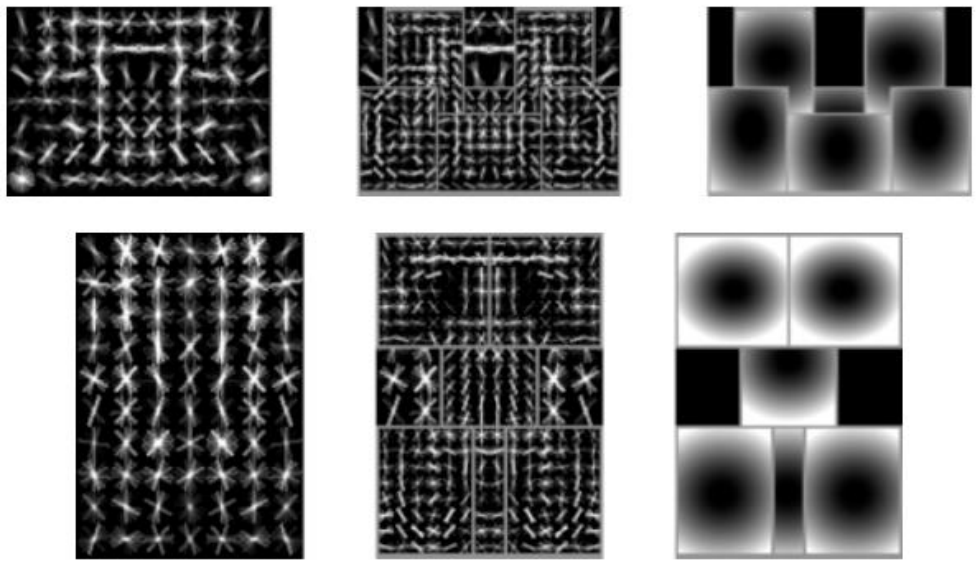
3. Hybrid template/parts model

Detections





Template Visualization
HoG features



root filters
coarse resolution

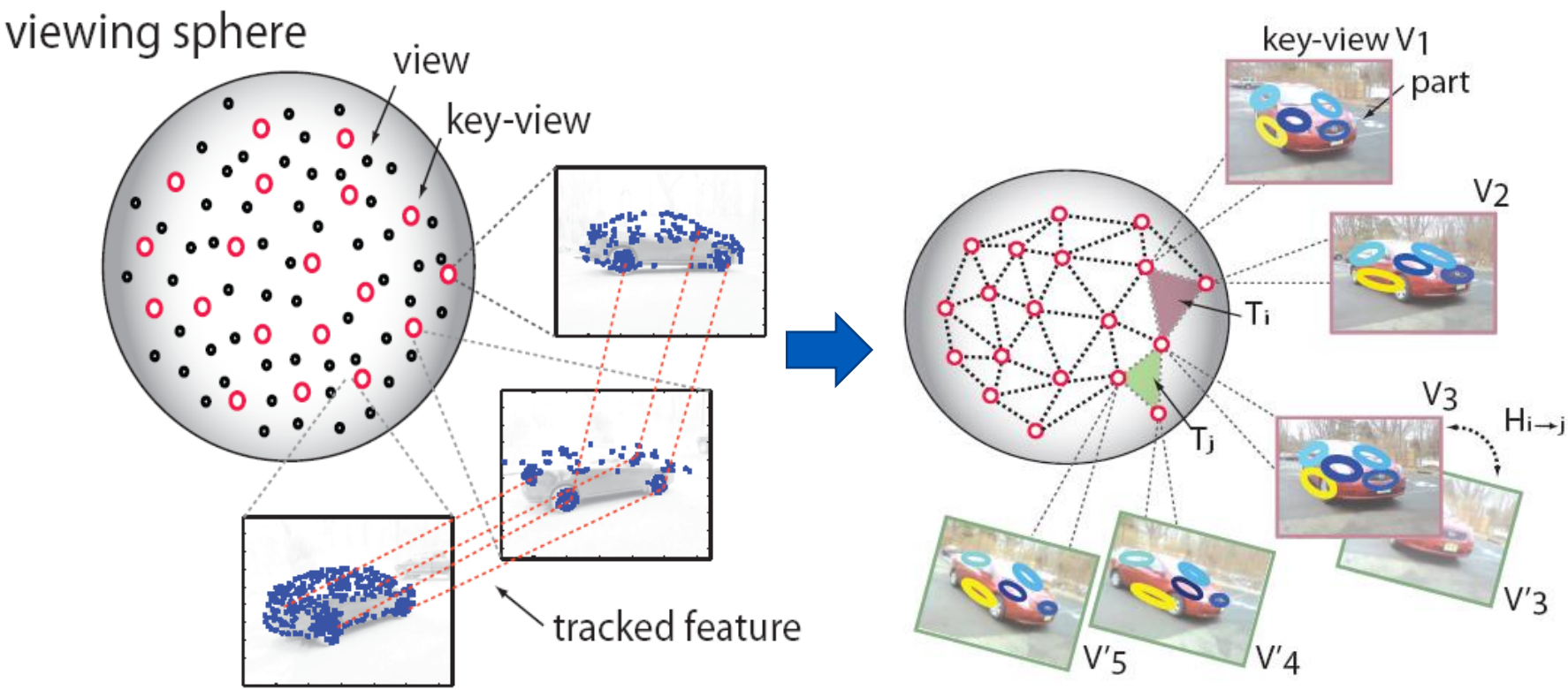
part filters
finer resolution

deformation
models

Felzenszwalb et al. 2008

# Specifying an object model

4. 3D-ish model
   - Object is collection of 3D planar patches under affine transformation



# General Process of Object Recognition

Specify Object Model

↓

Generate Hypotheses

Propose an alignment of the model to the image

↓

Score Hypotheses

↓

Resolve Detections

# Generating hypotheses

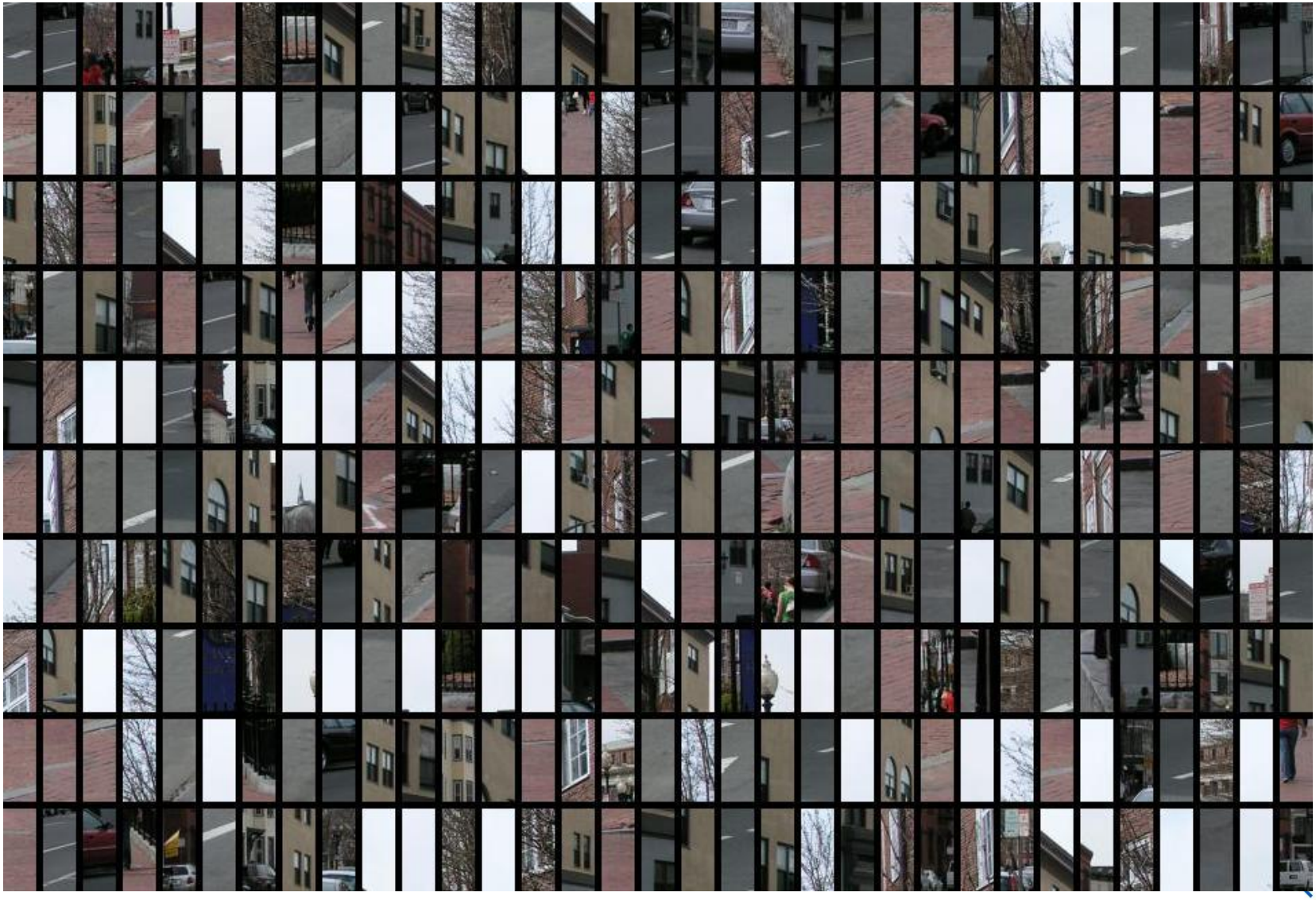
1. Sliding window
   - Test patch at each **location** and **scale**

# Generating hypotheses

1. Sliding window
   - Test patch at each **location** and **scale (pyramids)**

Note – Template did not change size

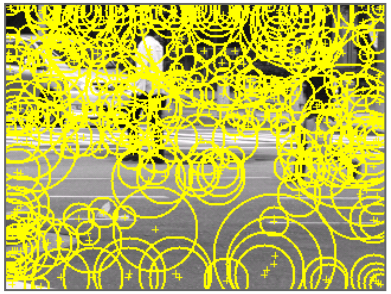
# Each window is separately **classified**



# Generating hypotheses

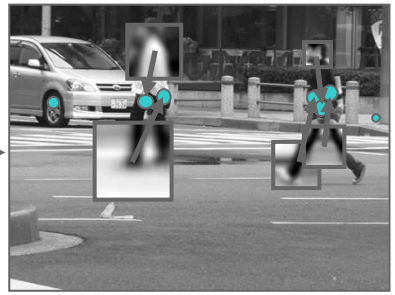
2. Voting from patches/keypoints

Interest Points

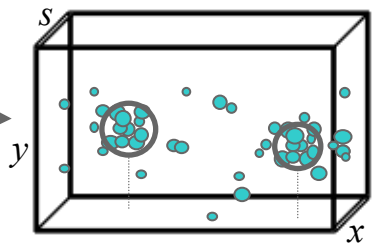


Matched Codebook Entries

Probabilistic Voting



3D Voting Space (continuous)



# Generating hypotheses

3. Region-based proposal

# General Process of Object Recognition

Specify Object Model

↓

Generate Hypotheses

↓

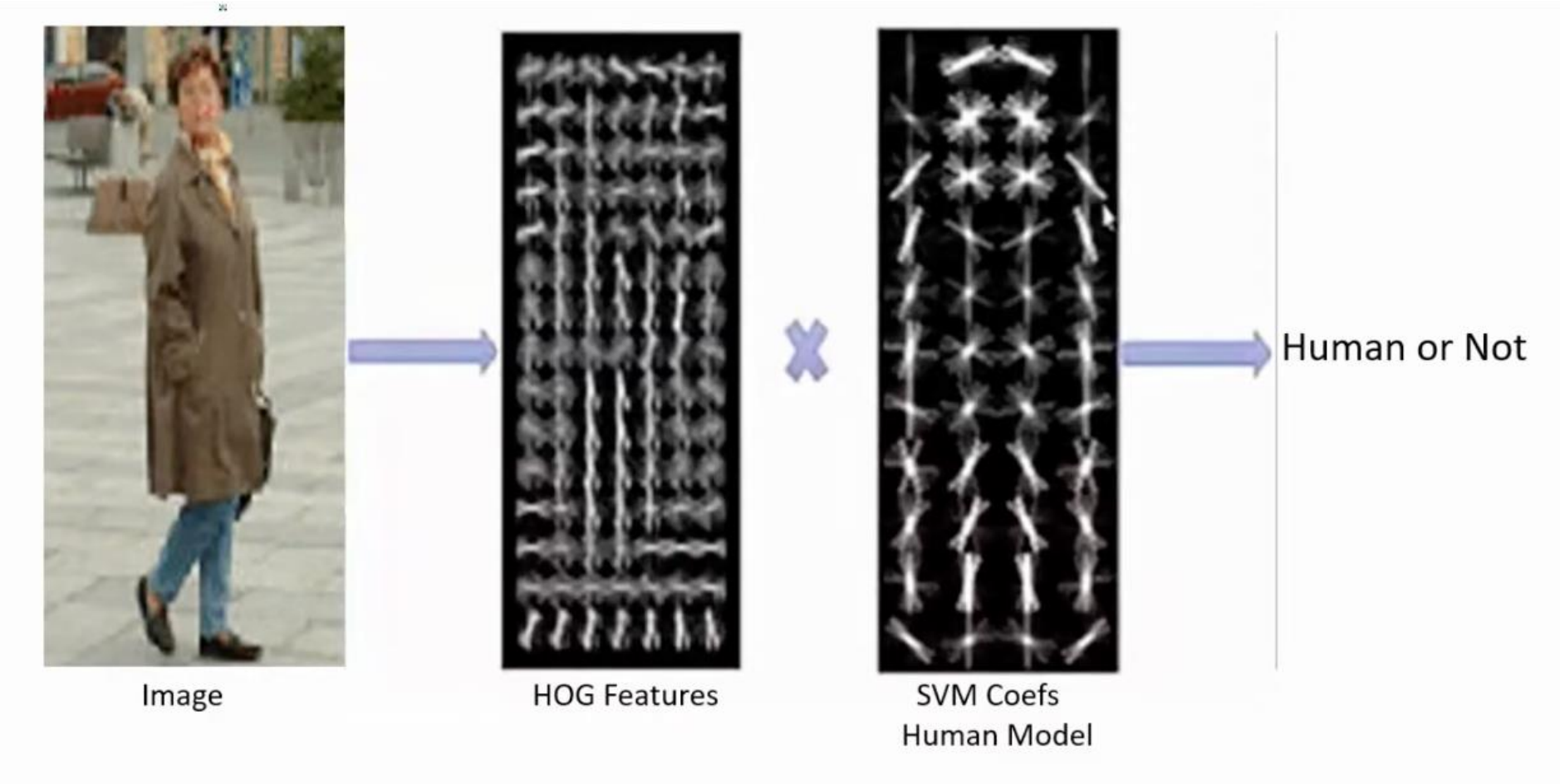
Score Hypotheses

↓

Resolve Detections

Mainly-gradient based features, usually based on summary representation, many classifiers.

# HOG + SVM for Object Detection




https://youtu.be/sDByI84n5mY



Histograms of Oriented Gradients for Human Detection, Navneet Dalal, Bill Triggs, International Conference on Computer Vision & Pattern Recognition, 2005

# General Process of Object Recognition

Specify Object Model

↓

Generate Hypotheses

↓

Score Hypotheses

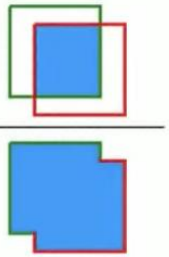
↓

Resolve Detections

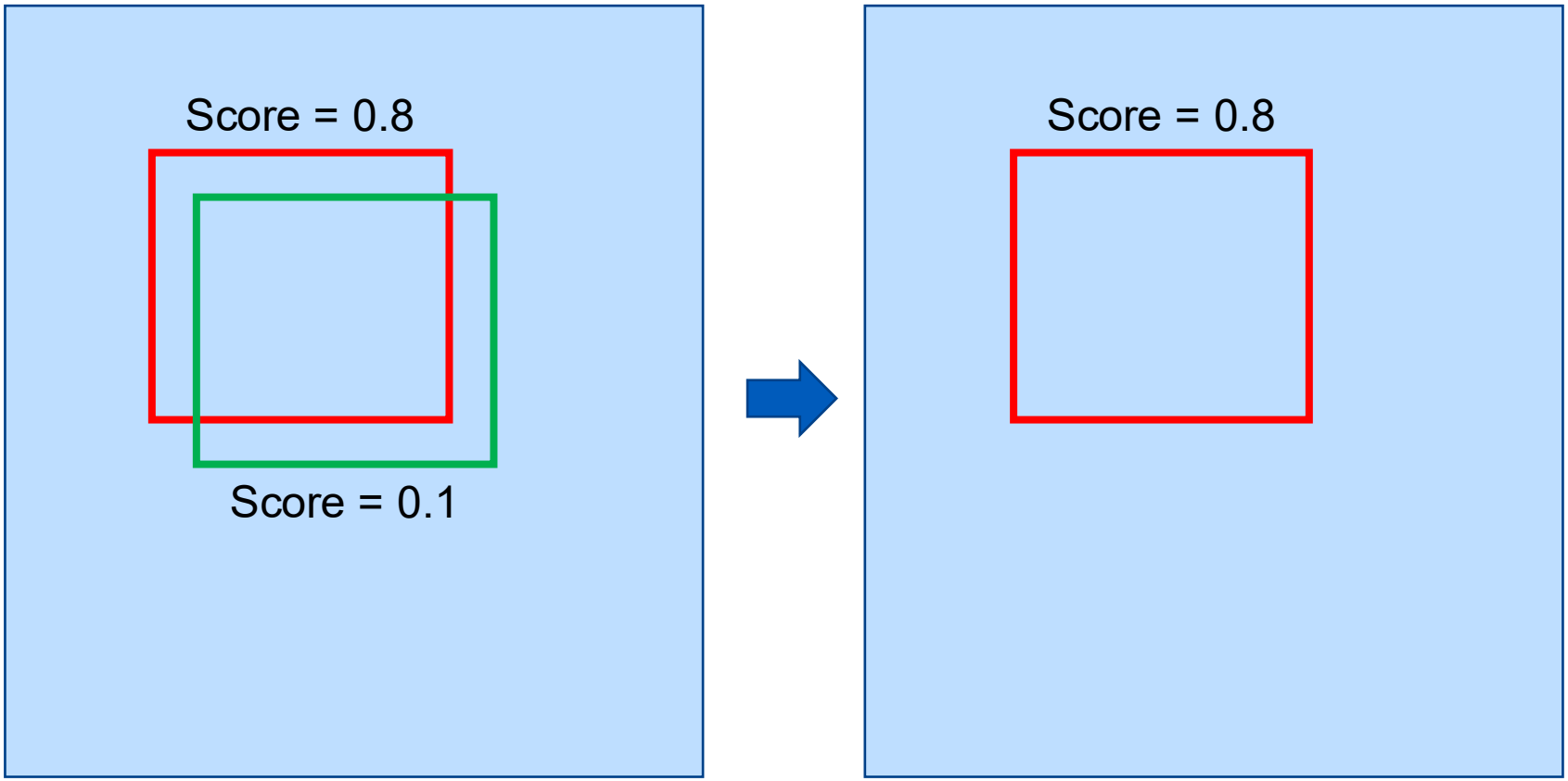
Rescore each proposed object based on whole set

# Resolving detection scores

$$IOU = \frac{\text{area of overlap}}{\text{area of union}} =$$

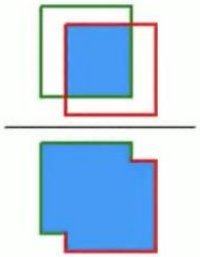


1. Non-max suppression

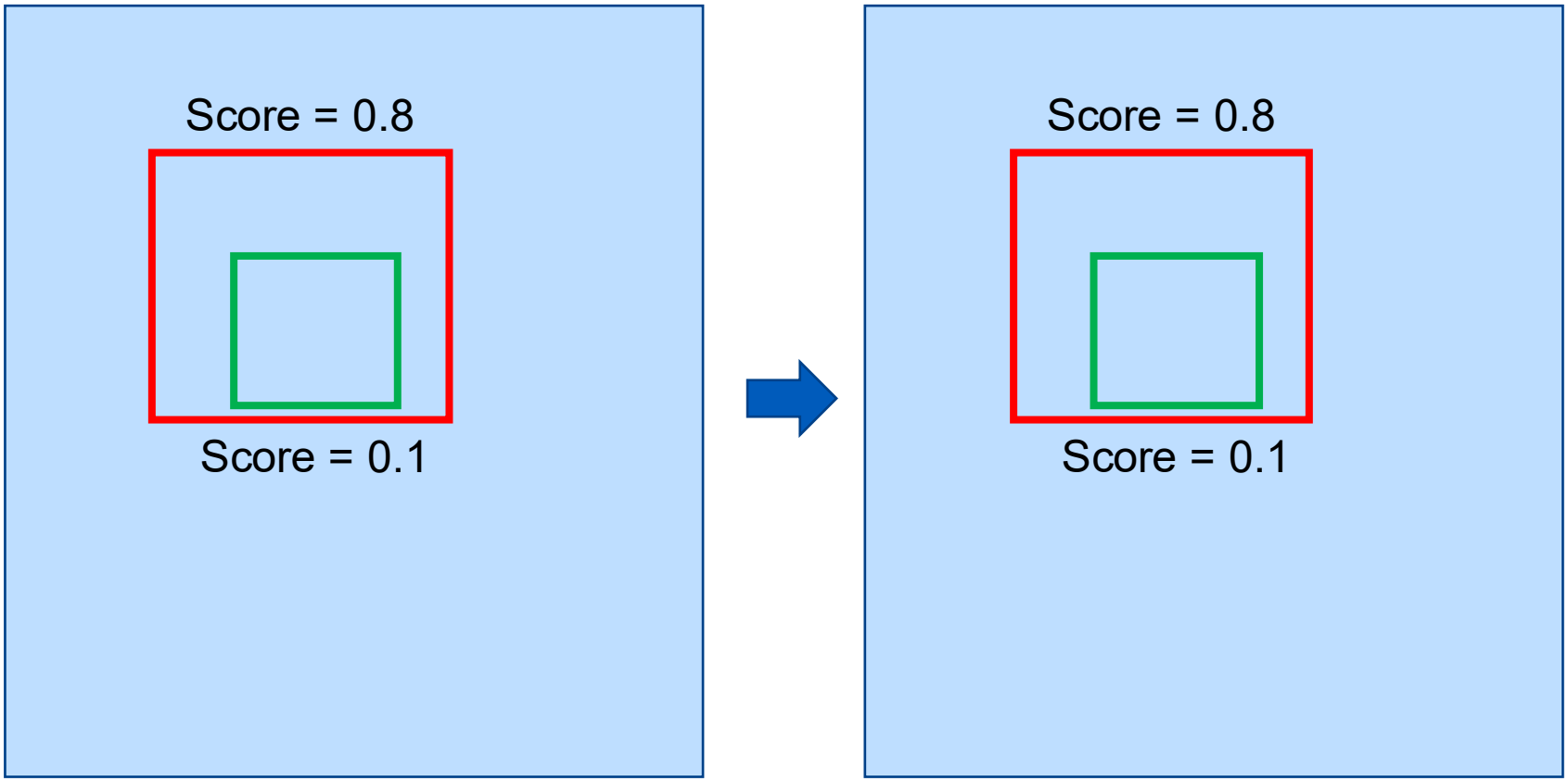


# Resolving detection scores

$$IOU = \frac{\text{area of overlap}}{\text{area of union}} =$$



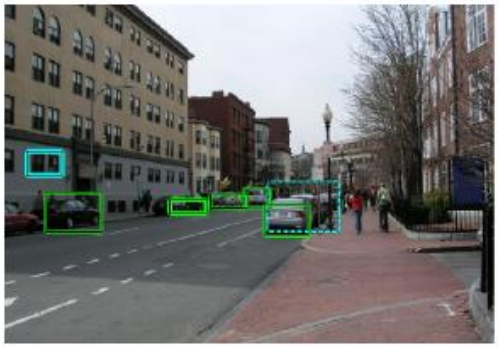
1. Non-max suppression




"Overlap" score is below some threshold

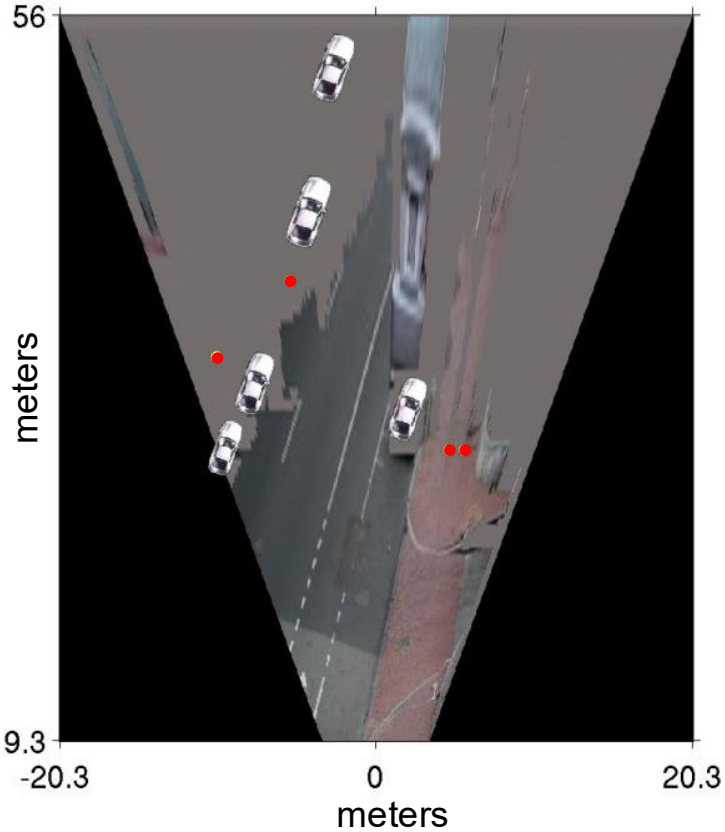
# Resolving Detection Scores

2. Context/reasoning



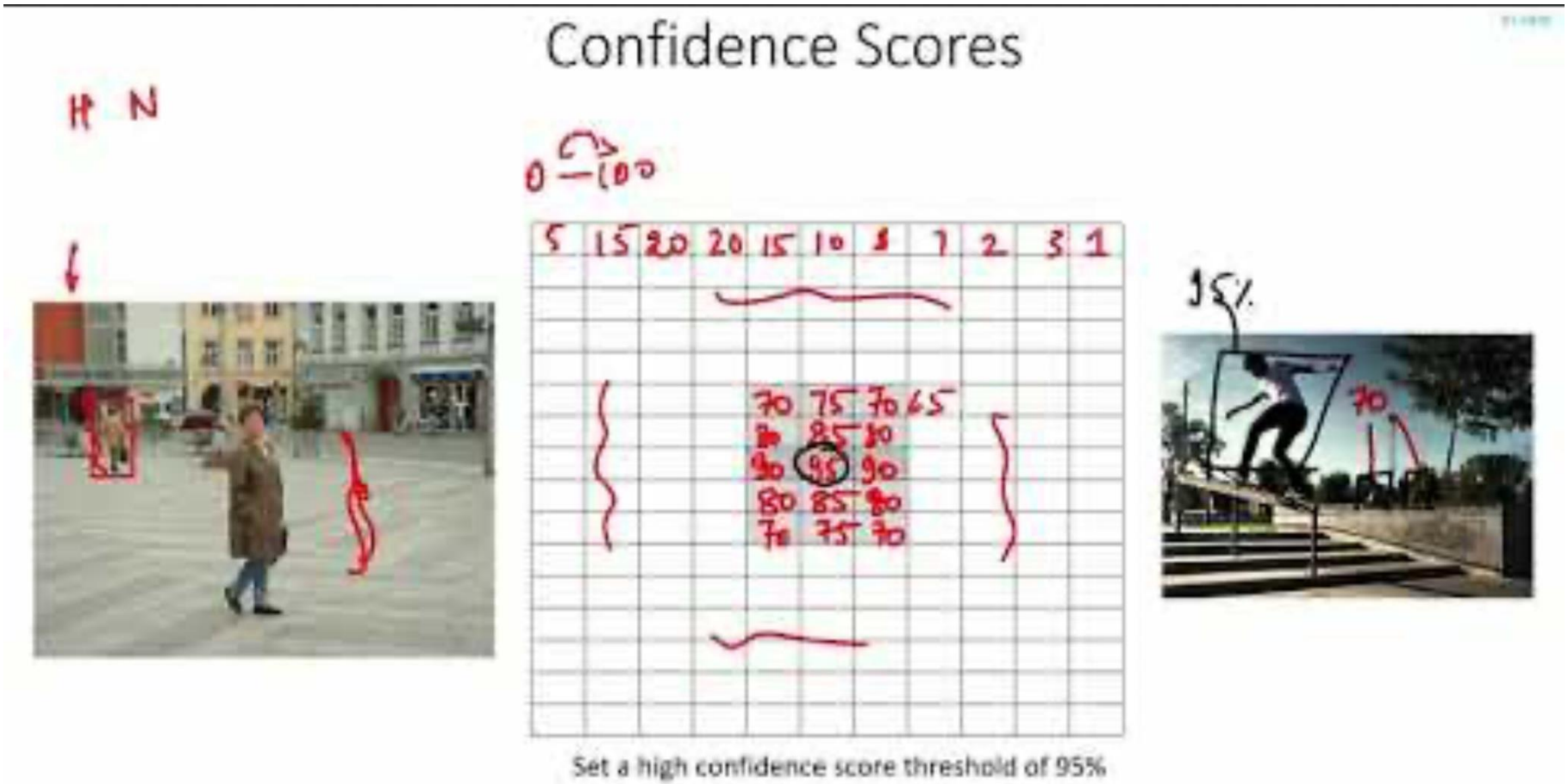
(g) Car Detections: Local

(h) Ped Detections: Local




Hoiem et al. 2006

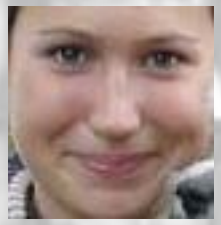
# Non-Max Suppression




https://youtu.be/07jFApuhh4I

# Face detection and recognition

Detection



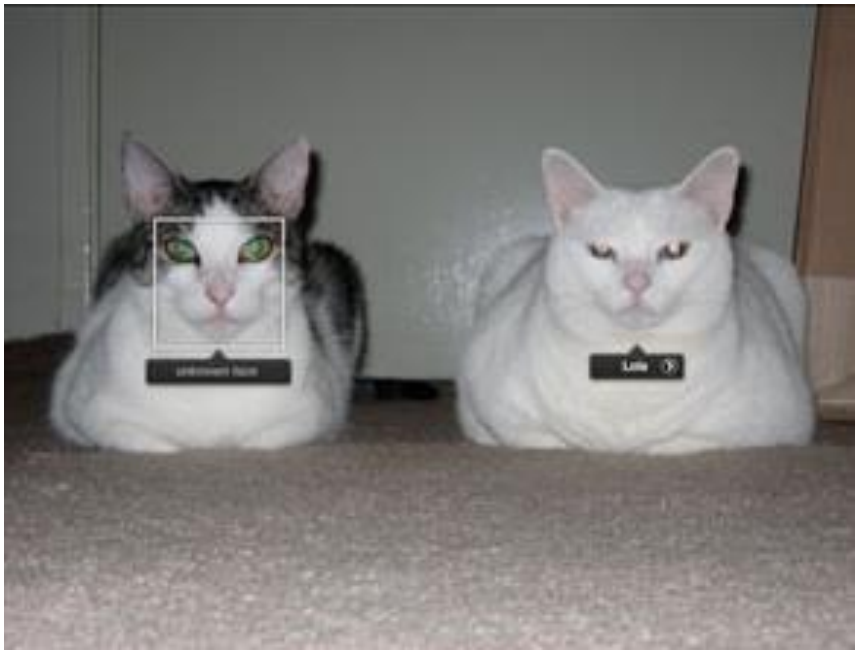
Recognition

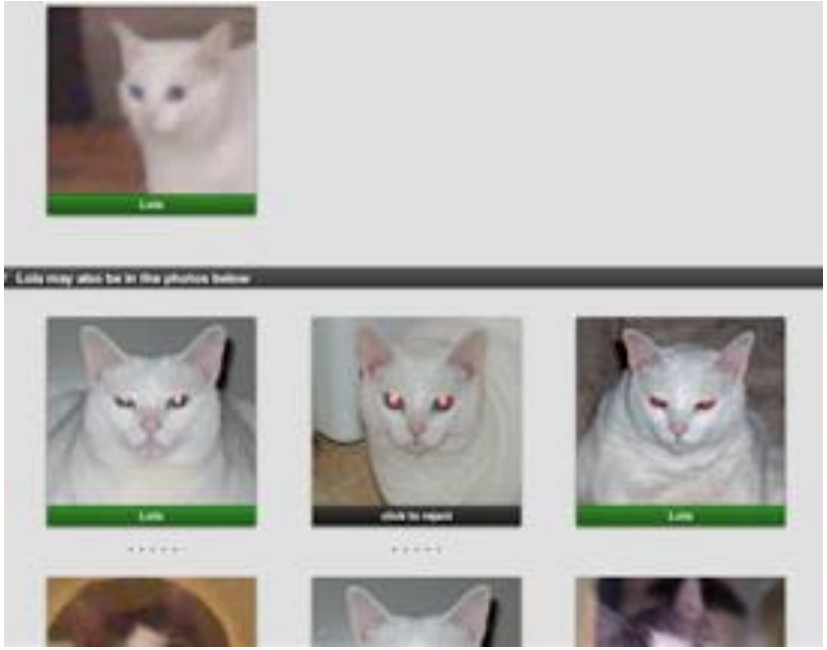
“Sally”

# Consumer application:

# Consumer application:

- Can be trained to recognize pets!





# Challenges of face detection

- Sliding window detector must evaluate tens of thousands of location/scale combinations
- Faces are rare: 0–10 per image
  - For efficiency, we should try to spend as little time as possible on the non-face windows
  - A megapixel image has ~$10^6$ pixels and a comparable number of candidate face locations
  - To avoid having a false positive in every image, our false positive rate has to be less than $1/10^6$

# The Viola/Jones Face Detector

- Viola-Jones Detection Framework
  - Sliding Window Face Detection
- A simple but effective approach to real-time object detection
- Training is slow, but detection is very fast
- Key ideas
  - *Integral images* for fast feature evaluation
  - *Boosting* for feature selection
  - *Attentional cascade* for fast rejection of non-face patches

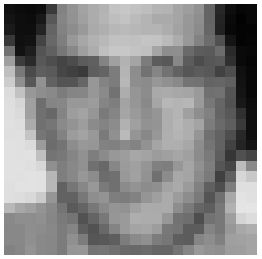


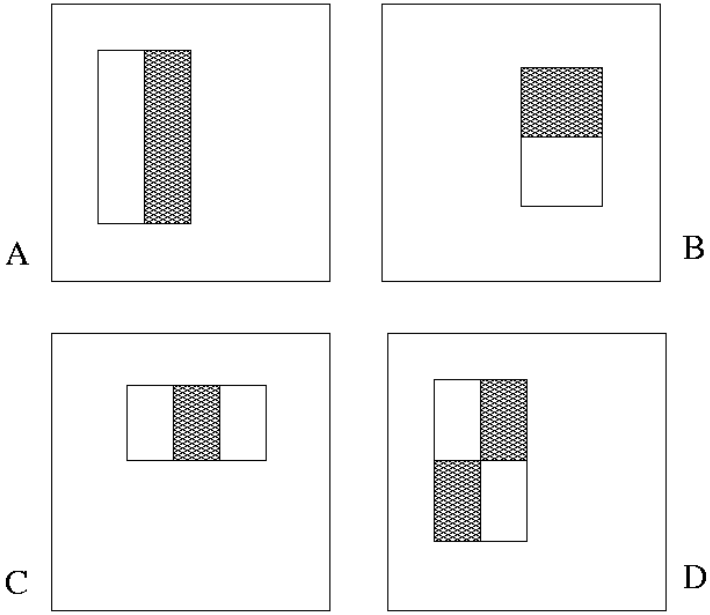
P. Viola and M. Jones. *Rapid object detection using a boosted cascade of simple features.* CVPR 2001.
P. Viola and M. Jones. *Robust real-time face detection.* IJCV 57(2), 2004.

# Image Features

- Simple Features that measures the difference in intensity

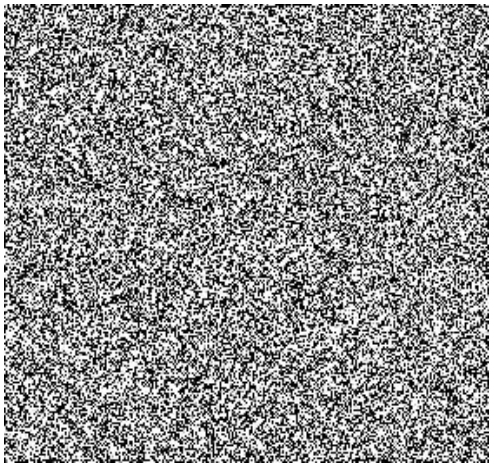
"Rectangle filters"





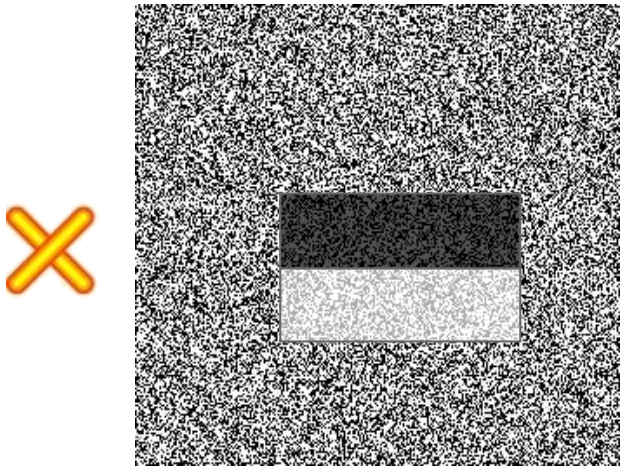

*Value = ∑ (pixels in white) – ∑ (pixels in black)*

# Example



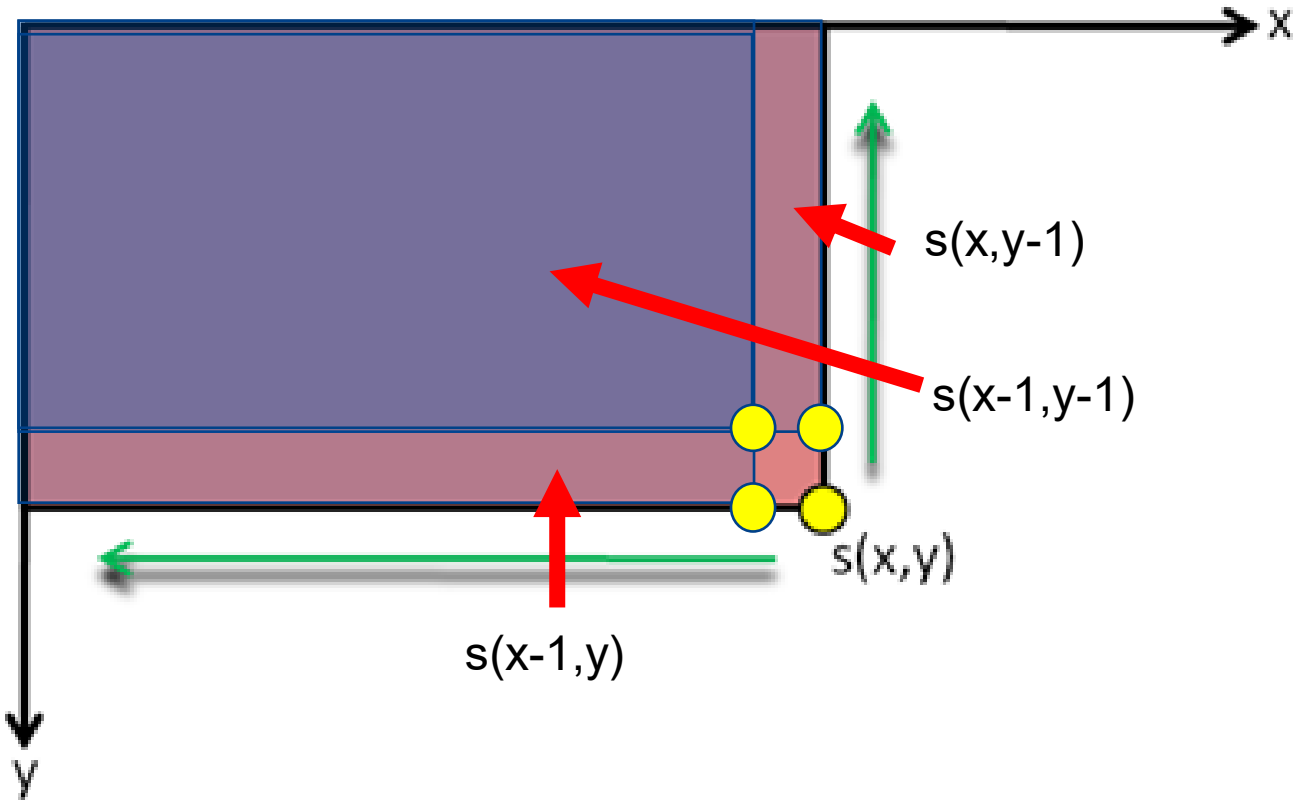
Source

Result



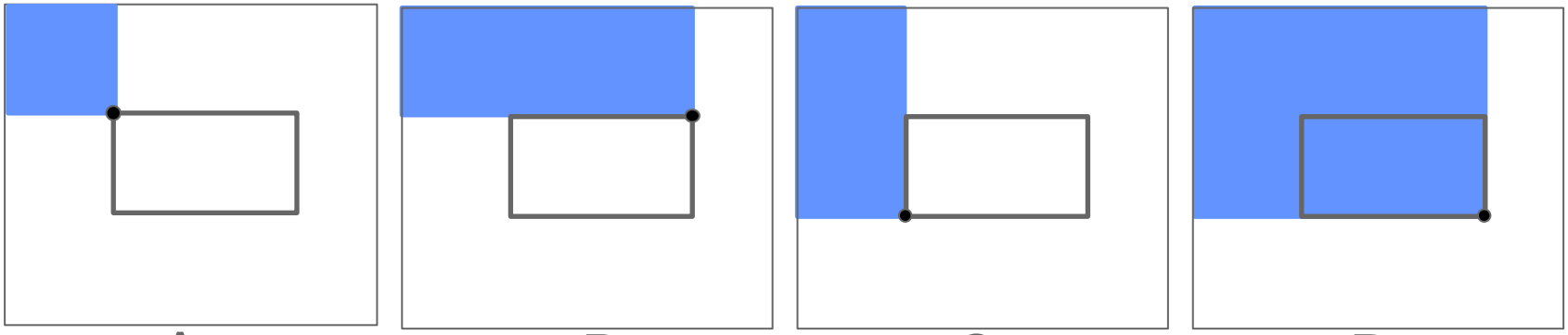
# Integral Image (Recap)

- A transformed image where every pixel is the sum of all pixels **above** and to the **left** of original image.



# Integral images (Recap)

- What's the sum of pixels in the Rectangle ABCD?



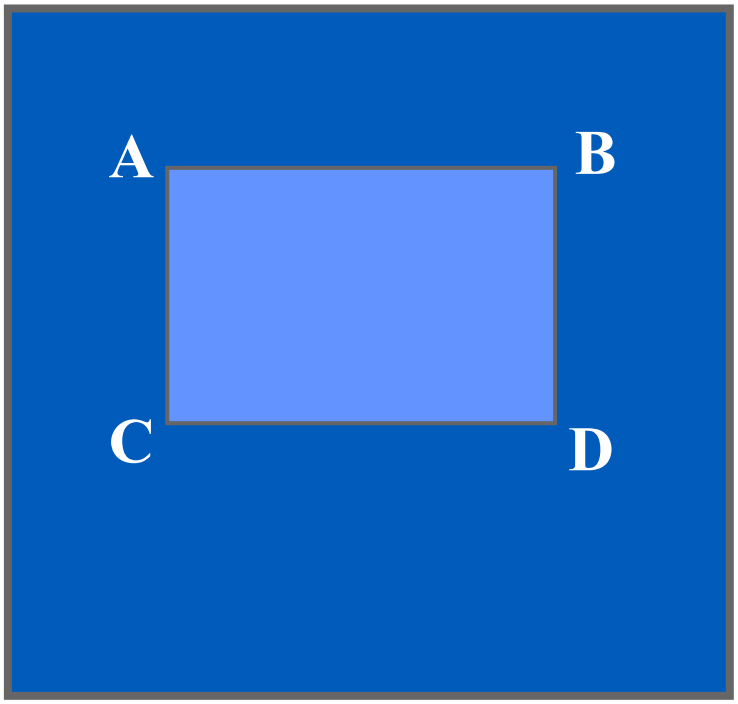
A B C D

# Computing sum within a rectangle (Recap)

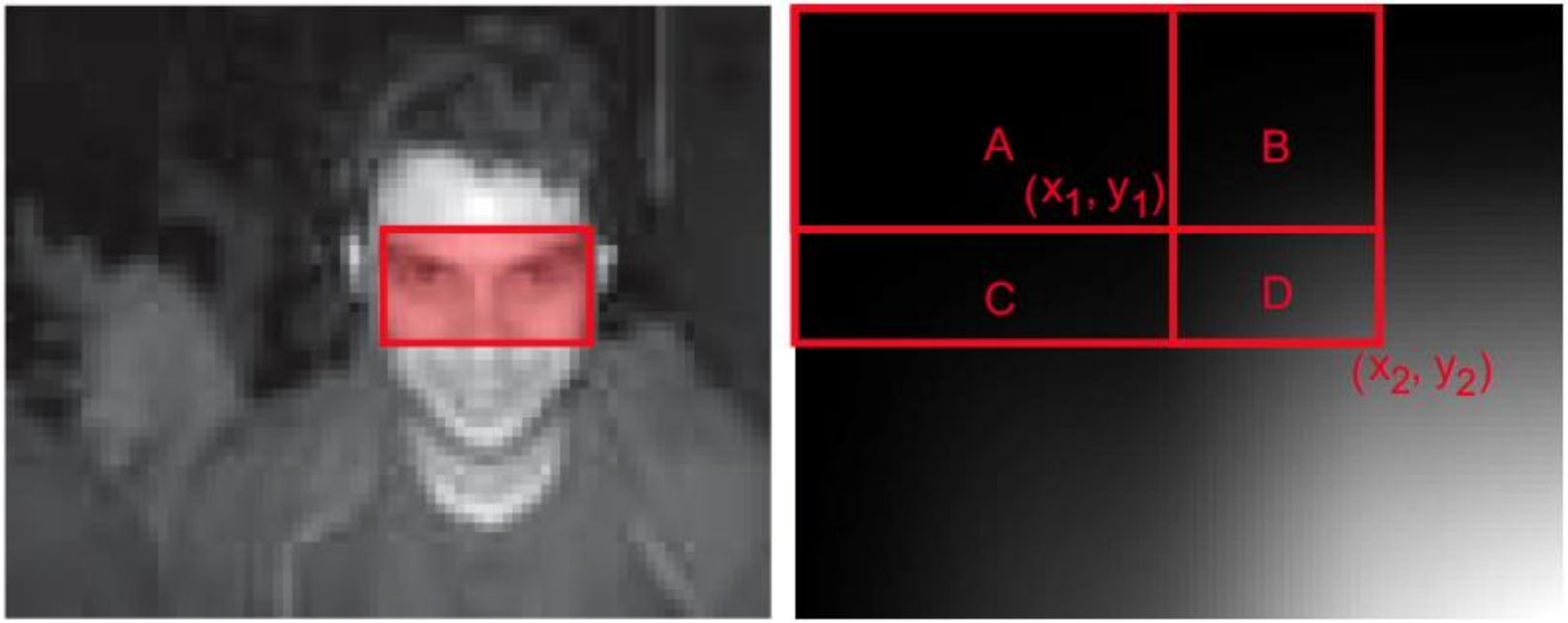
- Let A,B,C,D be the values of the integral image at the corners of a rectangle
- Then the sum of original image values within the rectangle can be computed as:

  sum = D – B – C + A

- Only 3 additions are required for any size of rectangle!

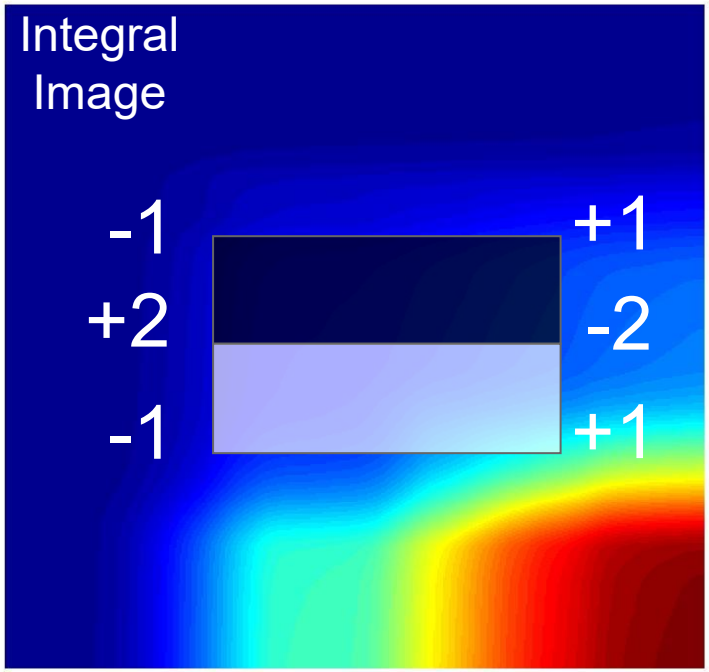


# Integral Image Example (Recap)



# Computing a rectangle feature

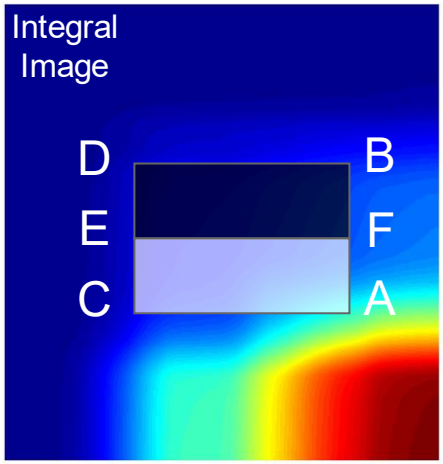
- Verify this:
  - Value = sum(scale factors * area of the regions).



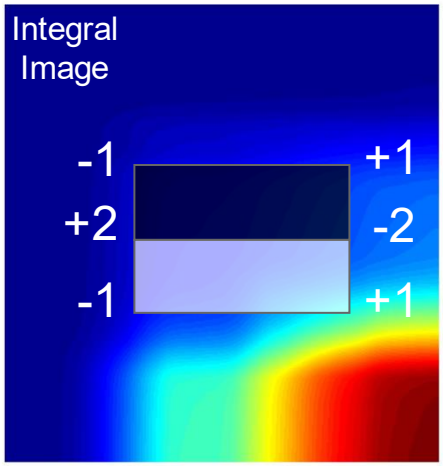

Exercise
10 minutes

# How do we get this?

- White Area – Black Area




- Sum of Entire Block = A – B – C + D
- White Block = A – F – C + E
- Black Block = F – E – B + D



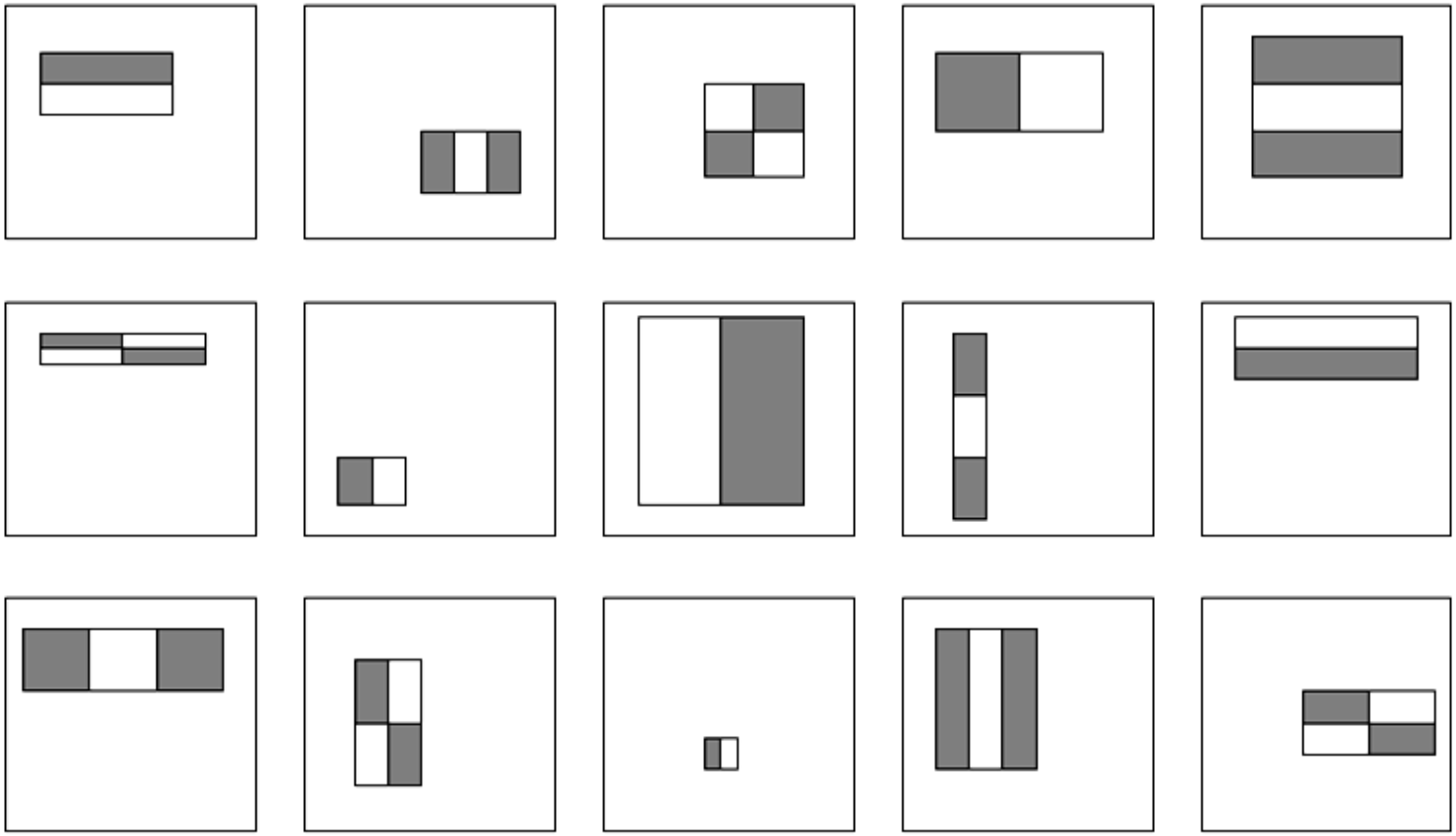

- White – Black =
  - A – 2F – C +2E - D + B

# Key ideas

- *Integral images* for **fast feature evaluation**
- ***Boosting*** for feature selection
- *Attentional cascade* for fast rejection of non-face windows

# Feature selection

- For a 24x24 detection region, the number of possible rectangle features is ~160,000!



# Feature selection

- At test time, it is impractical to evaluate the entire feature set
- Can we create a good classifier using just a small subset of all possible features?
- How to select such a subset?

# Boosting

- *Boosting* is a learning scheme that combines *weak learners* into a more accurate *ensemble classifier*
- Weak learners based on rectangle filters:

value of rectangle feature

$$h_t(x) = \begin{cases} 1 & \text{if } p_t f_t(x) > p_t \theta_t \\ 0 & \text{otherwise} \end{cases}$$

window

parity

threshold

- Ensemble classification function:

$$C(x) = \begin{cases} 1 & \text{if } \sum_{t=1}^{T} \alpha_t h_t(x) > \frac{1}{2} \sum_{t=1}^{T} \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

learned weights

A parity indicates the direction of the inequality sign

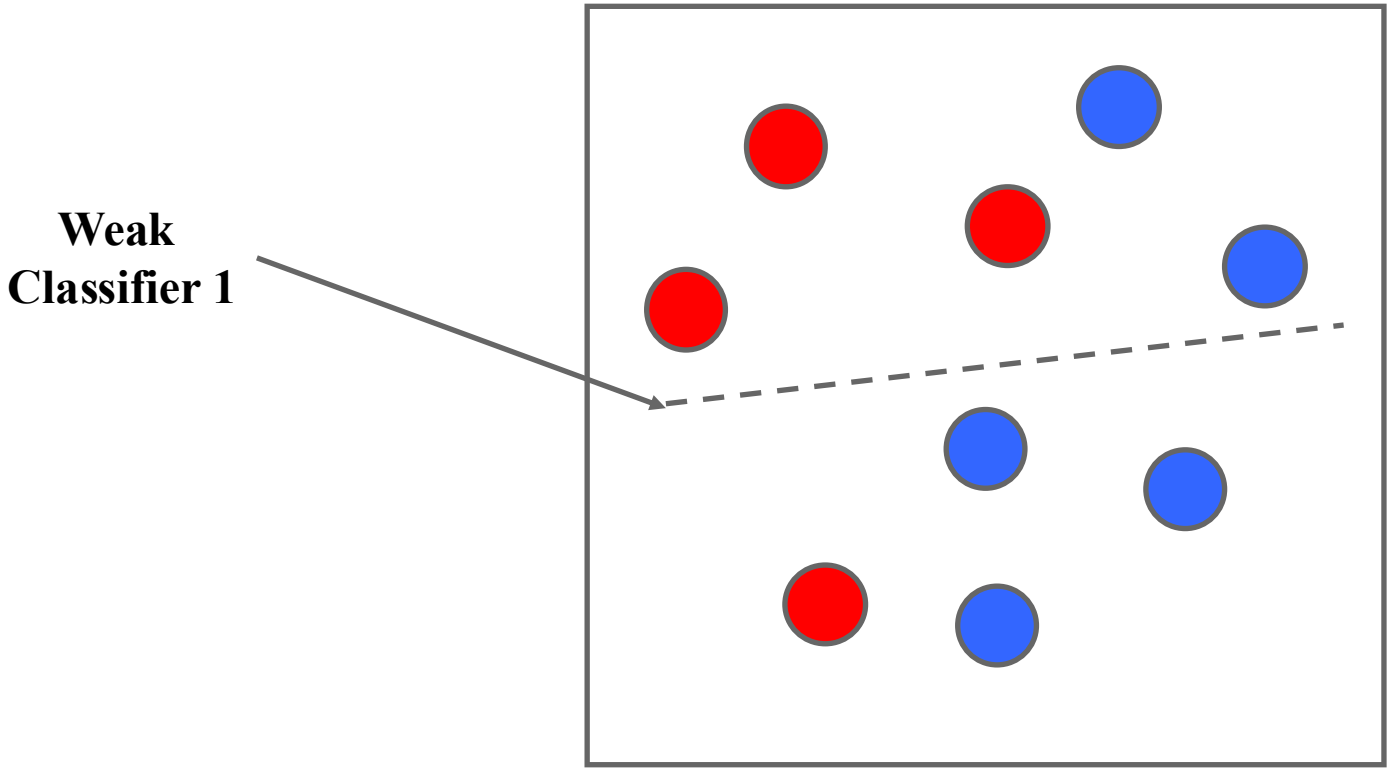
# Training procedure

- Initially, weight each training example equally
- In each boosting round:
  - Find the weak learner with lowest *weighted* training error
  - Raise the weights of training examples misclassified by current weak learner
- Compute final classifier as linear combination of all weak learners
  - Weight of each learner is proportional to its accuracy
  - Exact formulas for re-weighting and combining weak learners depend on the particular boosting scheme, i,e. Adaptive Boost (AdaBoost).



Y. Freund and R. Schapire, A short introduction to boosting, *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

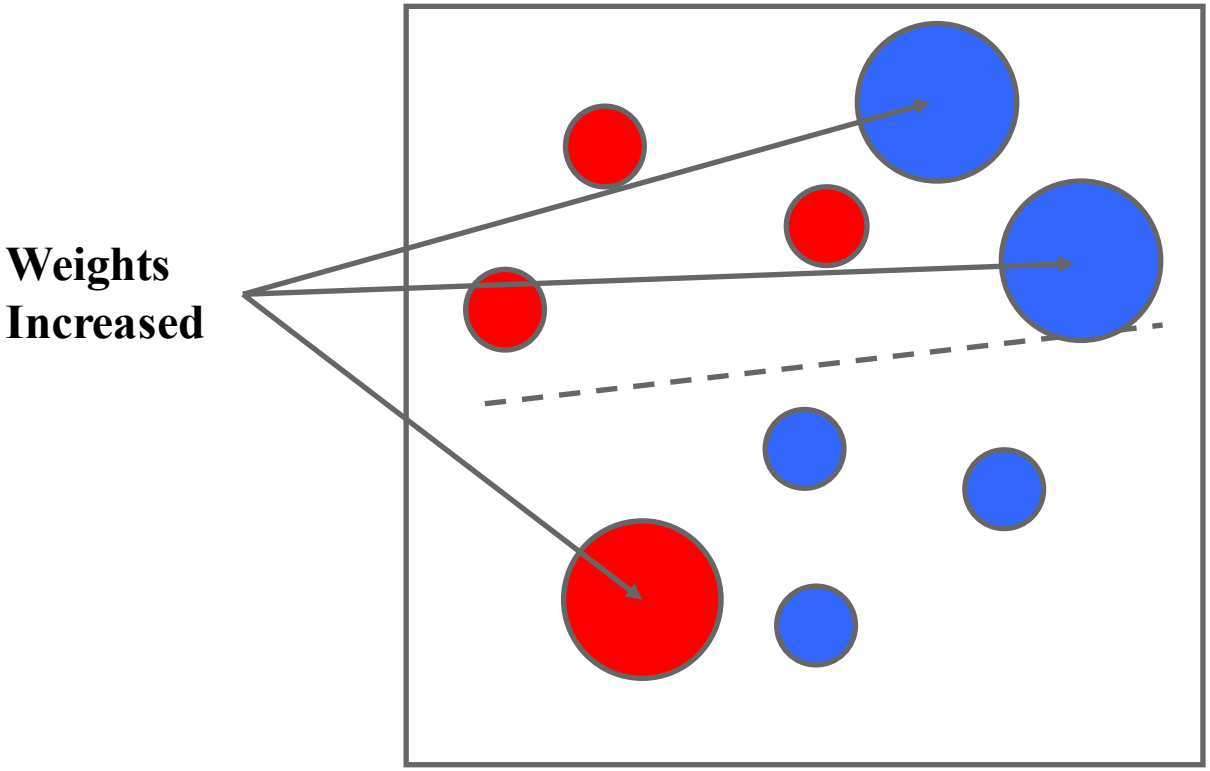
# Boosting intuition



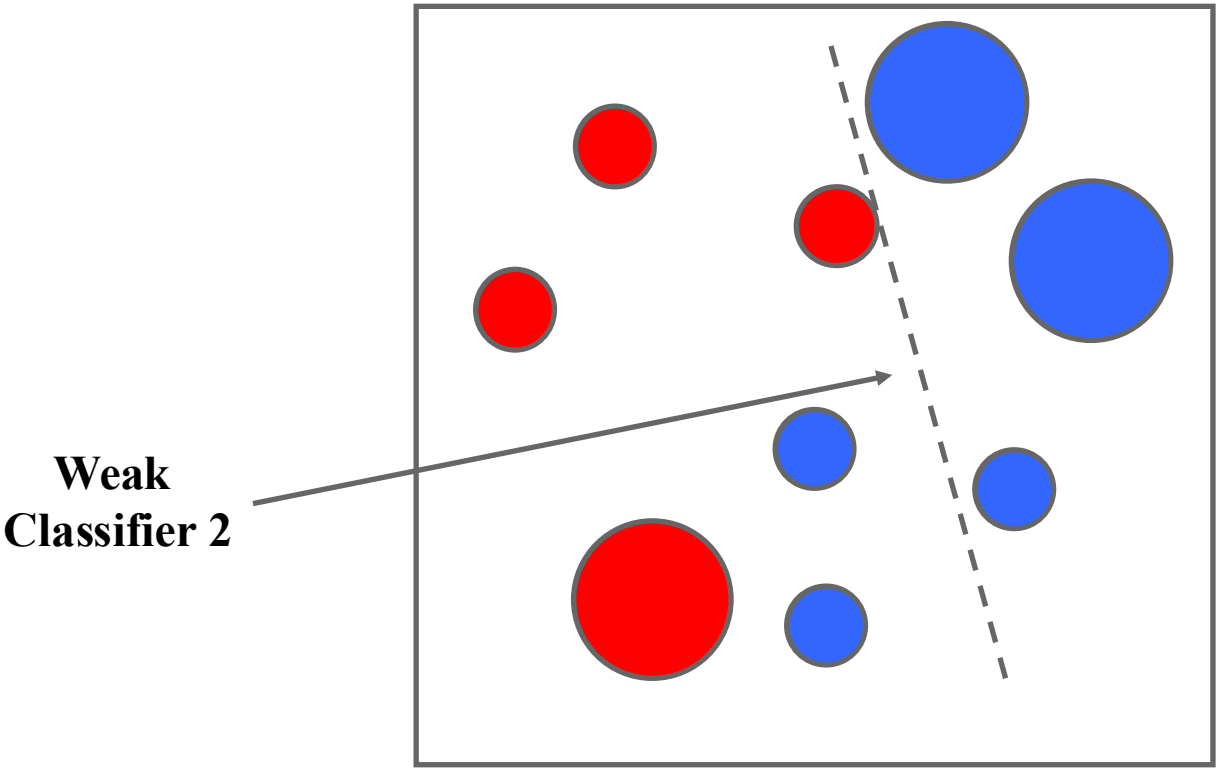



Slide credit: Paul Viola

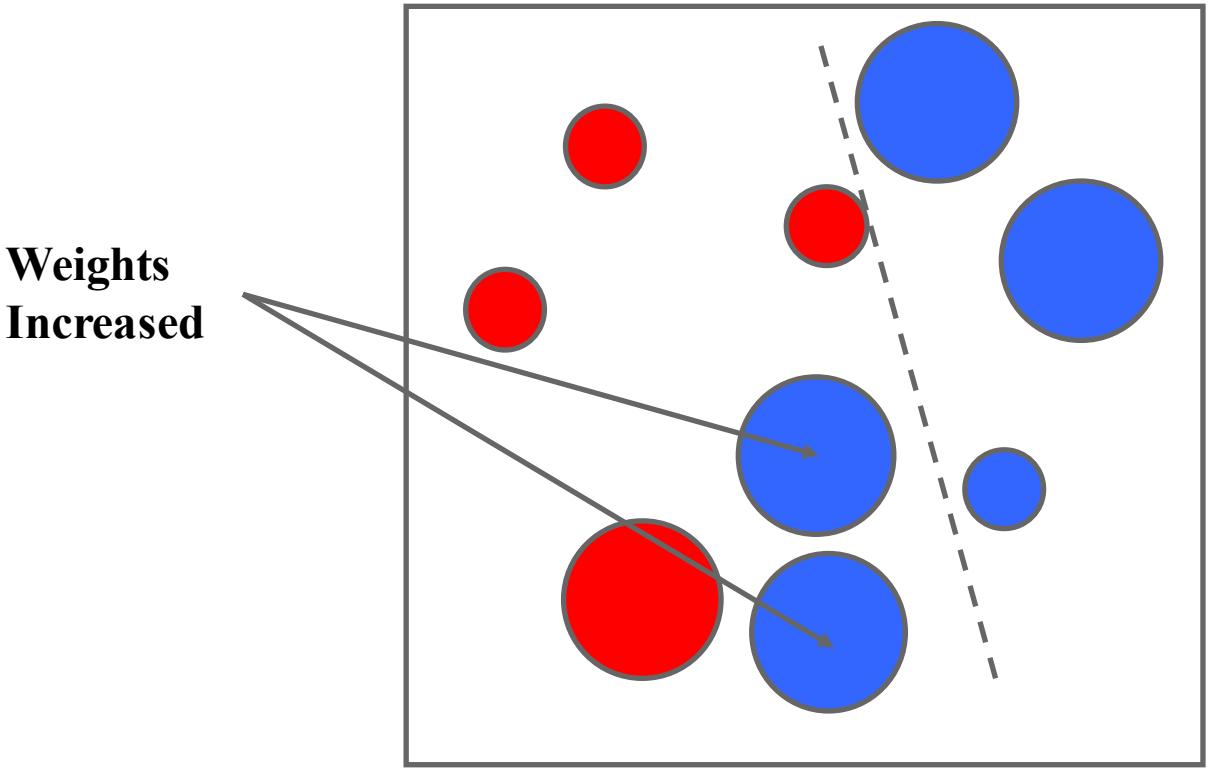
# Boosting illustration



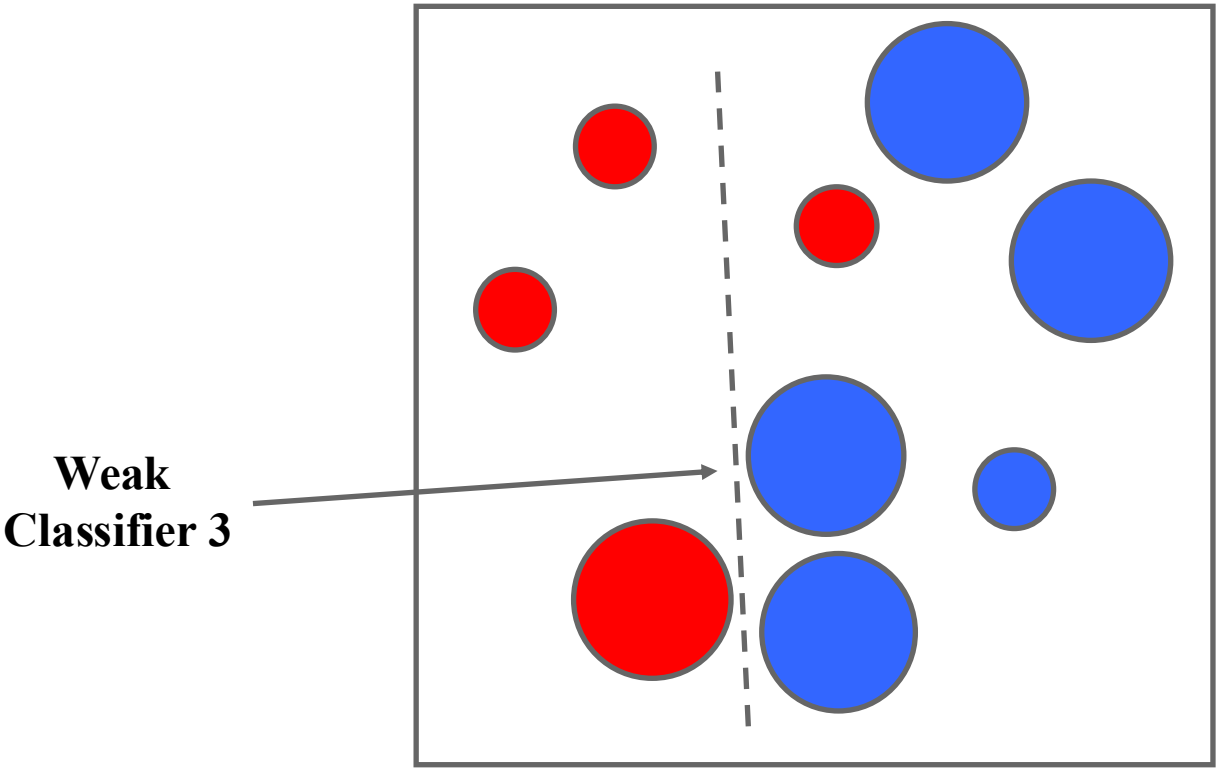
# Boosting illustration



# Boosting illustration

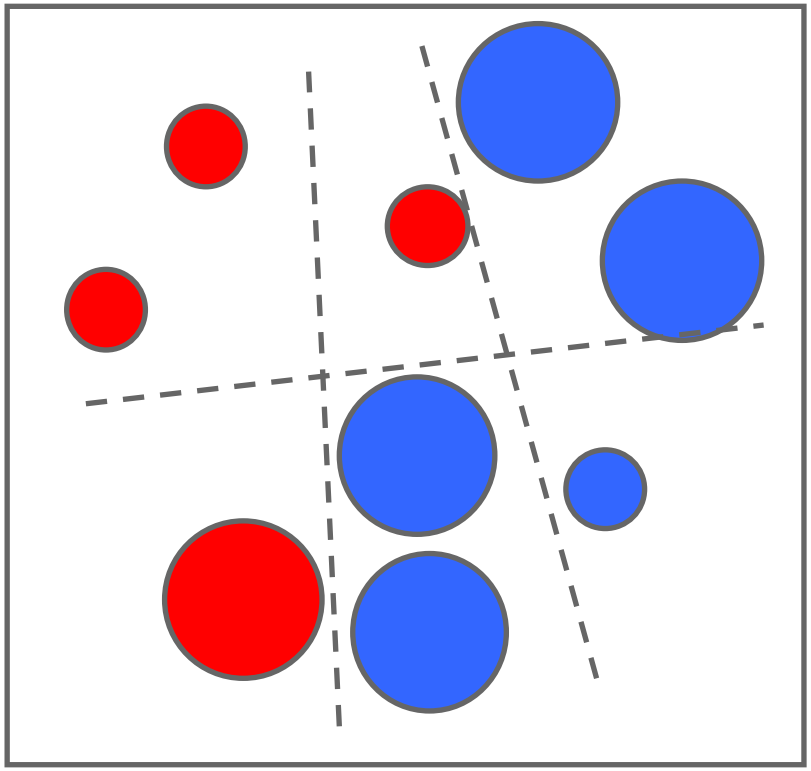


# Boosting illustration



# Boosting illustration

**Final classifier is a combination of weak classifiers**



# Training procedure

- Initially, weight each training example equally
- In each boosting round:
  - Find the weak learner with lowest *weighted* training error
  - Raise the weights of training examples misclassified by current weak learner
- Compute final classifier as linear combination of all weak learners
  - Weight of each learner is proportional to its accuracy
  - Exact formulas for re-weighting and combining weak learners depend on the particular boosting scheme, i,e. Adaptive Boost (AdaBoost)



Y. Freund and R. Schapire, A short introduction to boosting, *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

# Adaboost for face detection

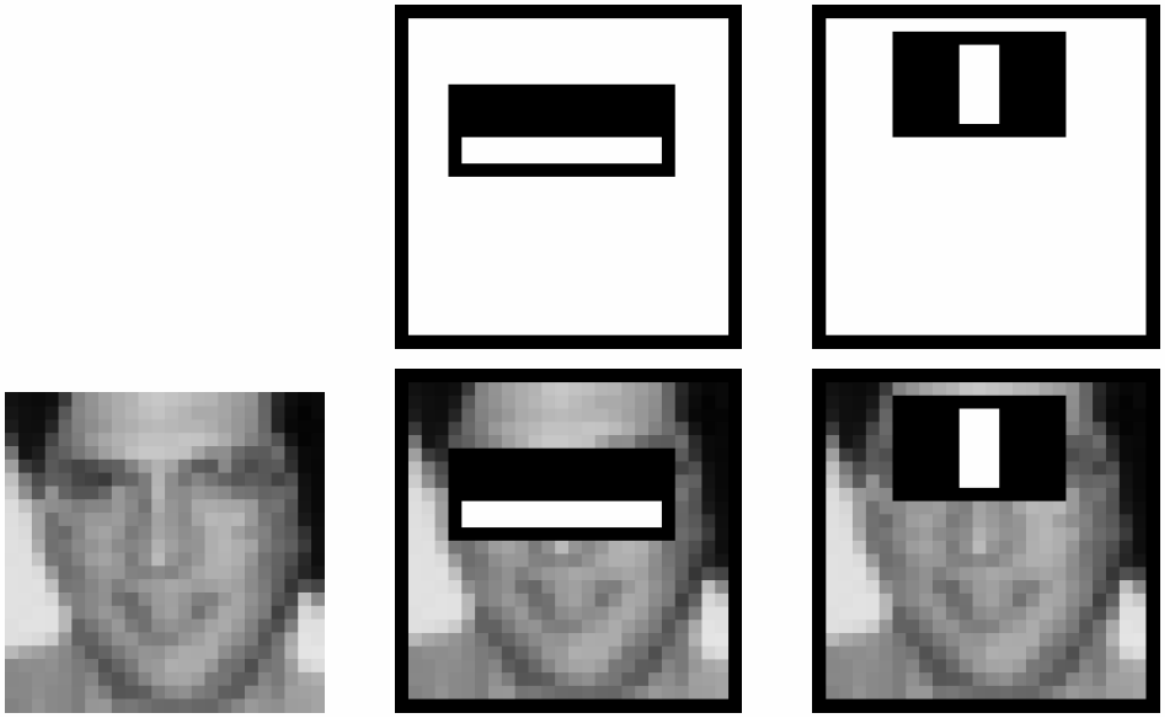
- Given example images $(x_1, y_1), \ldots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$ respectively, where $m$ and $l$ are the number of negatives and positives respectively.
- For $t = 1, \ldots, T$:
  1. Normalize the weights,
  $$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^{n} w_{t,j}}$$
  so that $w_t$ is a probability distribution.
  2. For each feature, $j$, train a classifier $h_j$ which is restricted to using a single feature. The error is evaluated with respect to $w_t$, $\epsilon_j = \sum_i w_i \left| h_j(x_i) - y_i \right|$.
  3. Choose the classifier, $h_t$, with the lowest error $\epsilon_t$.
  4. Update the weights:
  $$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$
  where $e_i = 0$ if example $x_i$ is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.
- The final strong classifier is:
$$h(x) = \begin{cases} 1 & \sum_{t=1}^{T} \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^{T} \alpha_t \\ 0 & \text{otherwise} \end{cases}$$
where $\alpha_t = \log \frac{1}{\beta_t}$



P. Viola and M. Jones. *Rapid object detection using a boosted cascade of simple features.* CVPR 2001.

# Boosting for face detection

- First two features selected by boosting:
- This feature combination can yield 100% recall and 50% false positive rate



# Boosting vs. SVM

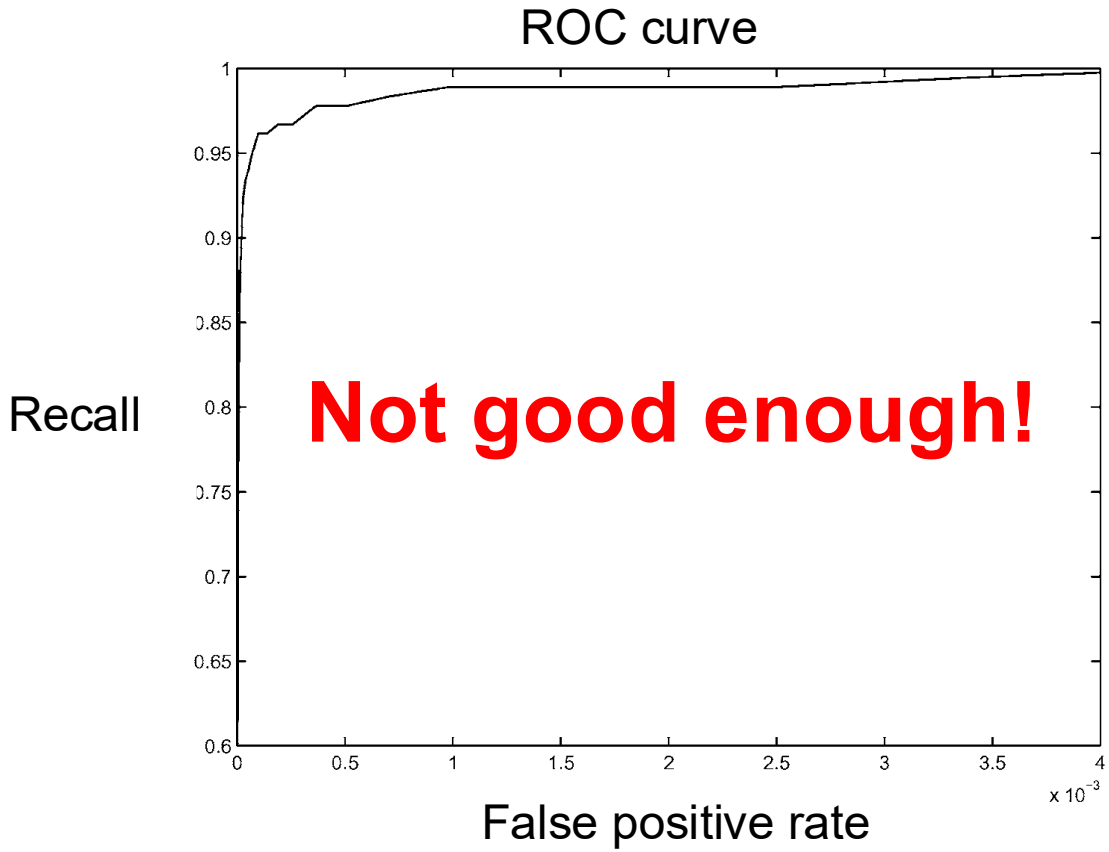
- Advantages of boosting
  - Integrates classifier training with feature selection
  - Complexity of training is linear instead of quadratic in the number of training examples
  - Flexibility in the choice of weak learners, boosting scheme
  - Testing is fast
- Disadvantages
  - Needs many training examples
  - Training is slow
  - Many-class problems doesn't work as well as SVM

# Key ideas

- *Integral images* for **fast feature evaluation**
- *Boosting* for feature selection
- *Attentional* ***cascade*** for fast rejection of non-face windows

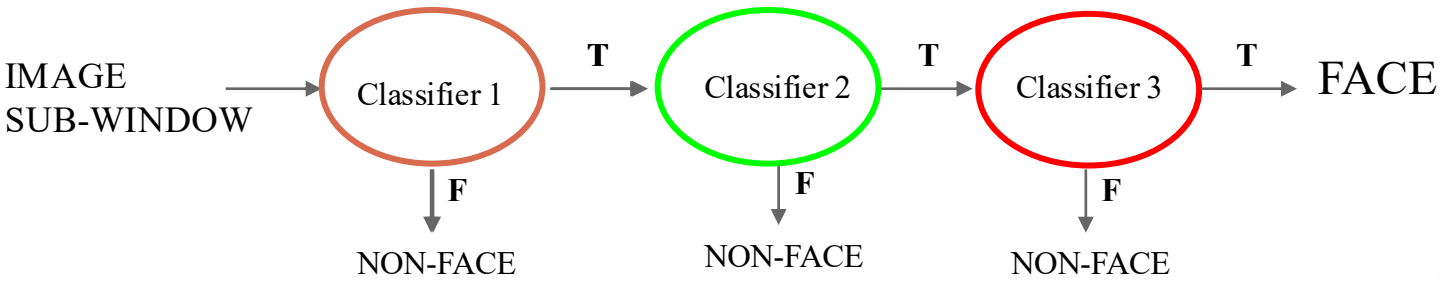
# Boosting for face detection

- A 200-feature classifier can yield 95% detection rate and a false positive rate of 1/14084



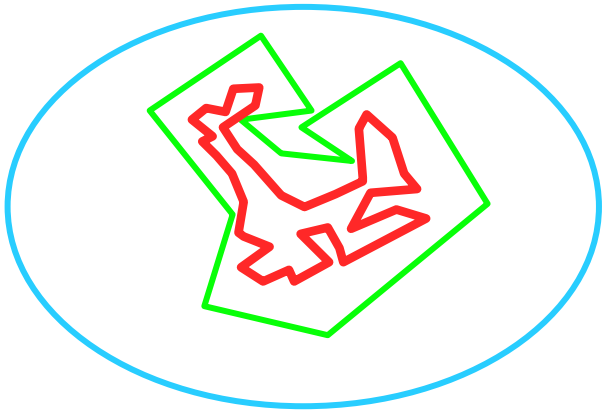
# Attentional cascade

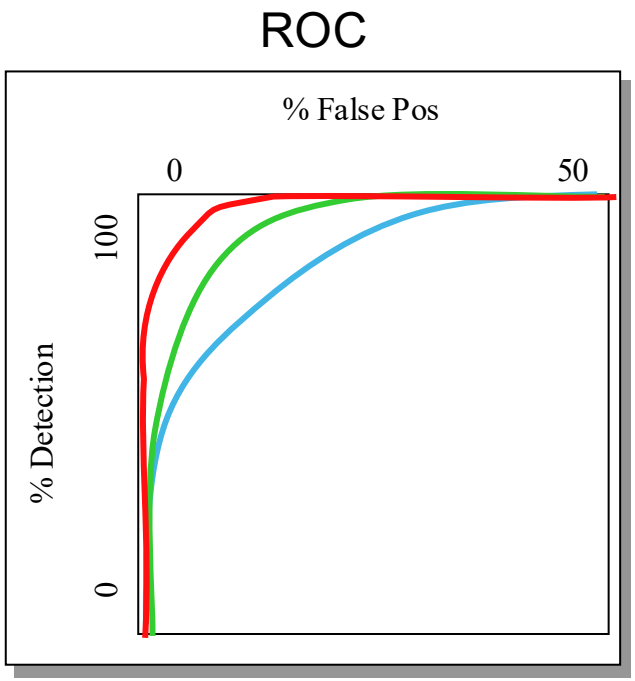
- Start with simple classifiers which reject many negatives while detecting almost all positives.
- Positive response from the first classifier triggers the evaluation of a second (more complex) classifier, etc.
- A negative outcome at any point leads to the immediate rejection of the sub-window

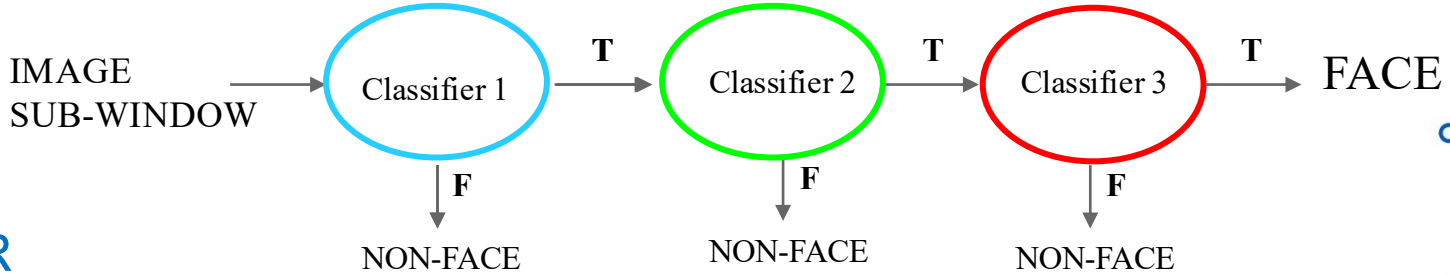


# Attentional cascade

- Chain classifiers that are progressively more complex and have lower false positive rates

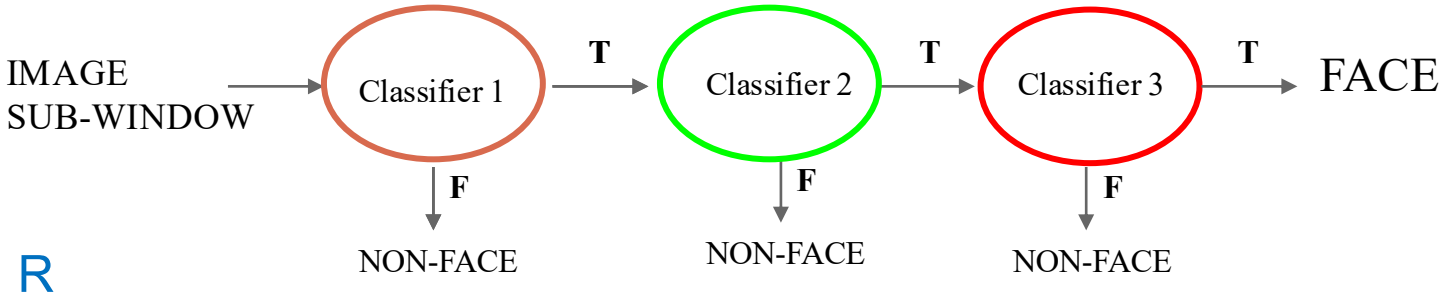




# Attentional cascade

- The detection rate and the false positive rate of the cascade are found by multiplying the respective rates of the individual stages
- A detection rate of 0.9 and a false positive rate $10^{-6}$ can be achieved by a 10-stage cascade if each stage has a detection rate of 0.99 ($0.99^{10} \approx 0.9$) and a false positive rate of about 0.30 ($0.3^{10} \approx 6\times10^{-6}$)



# Training the cascade

- Set target detection and false positive rates for each stage
- Keep adding features to current stage until target rates met
  - Lower AdaBoost threshold to maximize detection
    - opposed to minimizing total classification error
  - Test on a *validation set*
- If the overall false positive rate is not low enough
  - Add another stage
- Use false positives from current stage as the negative training examples for the next stage

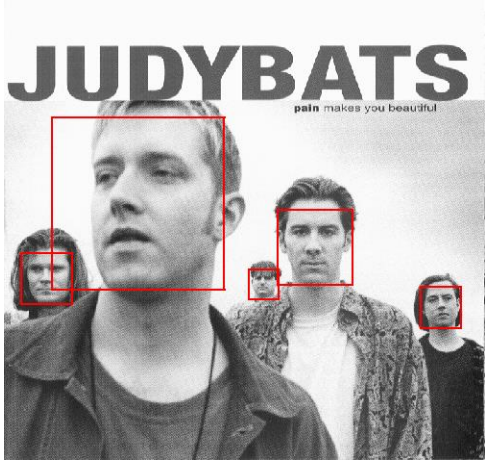
# The implemented system

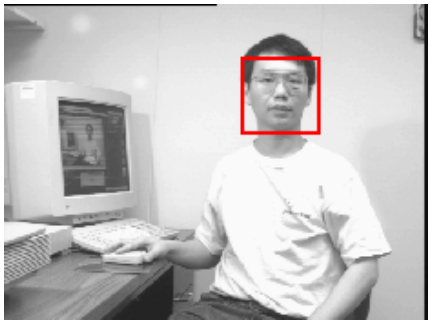
- Training Data
  - 5000 faces
    - All frontal, rescaled to 24x24 pixels
  - 300 million non-faces
    - 9500 non-face images
  - Faces are normalized
    - Scale, translation
- Many variations
  - Across individuals
  - Illumination
  - Pose

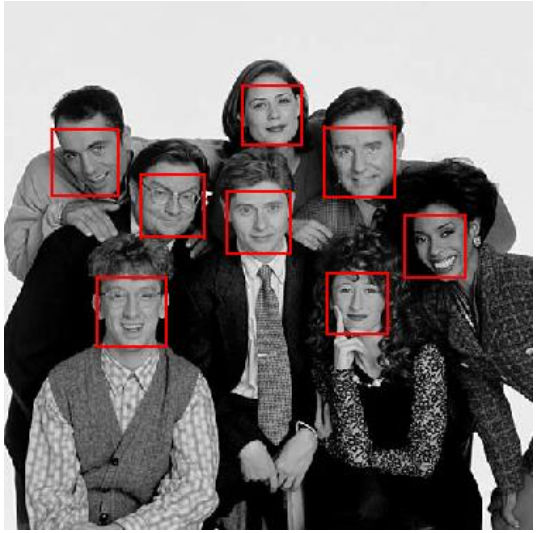
# System performance

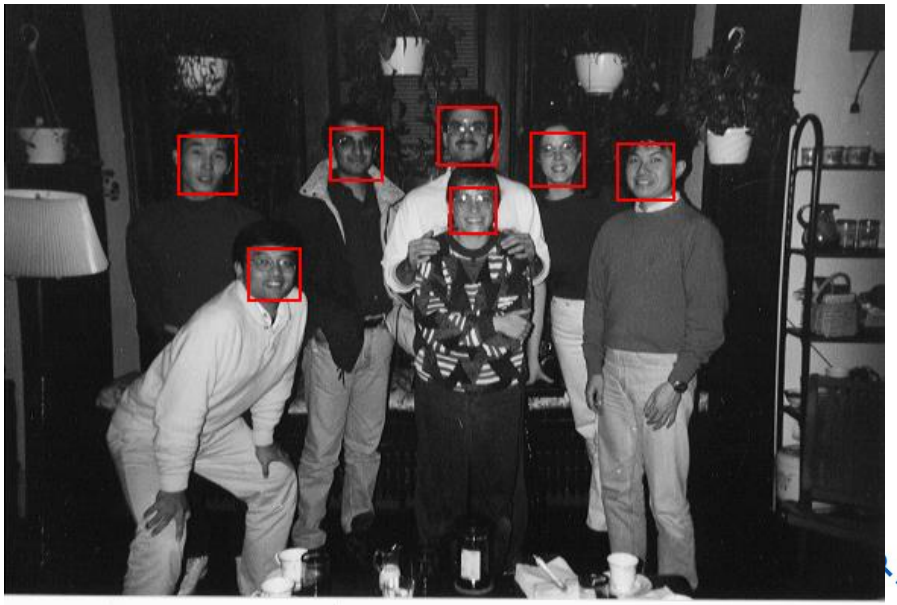
- Training time: “weeks” on 466 MHz Sun workstation
- 38 layers, total of 6061 features
- Average of 10 features evaluated per window on test set
- “On a 700 Mhz Pentium III processor, the face detector can process a 384 by 288 pixel image in about .067 seconds”
  - 15 Hz
  - 15 times faster than previous detector of comparable accuracy (Rowley et al.,1998)

# Output of Face Detector on Test Images



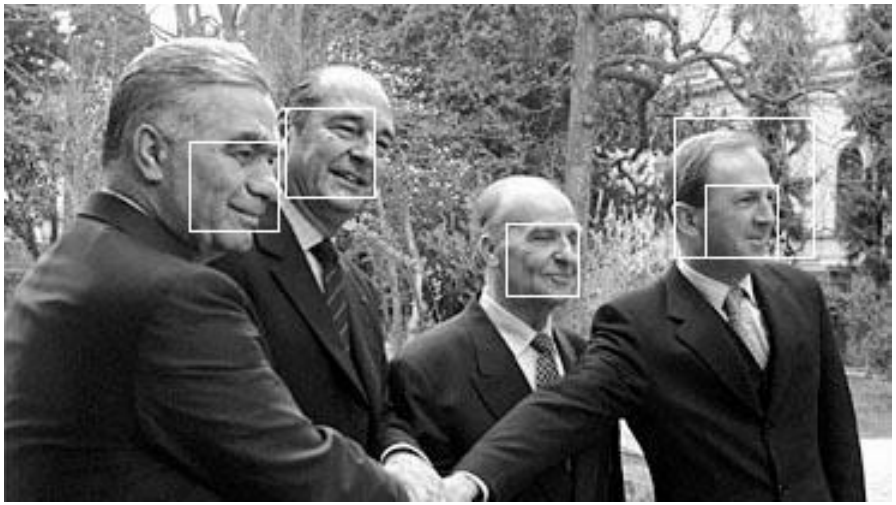







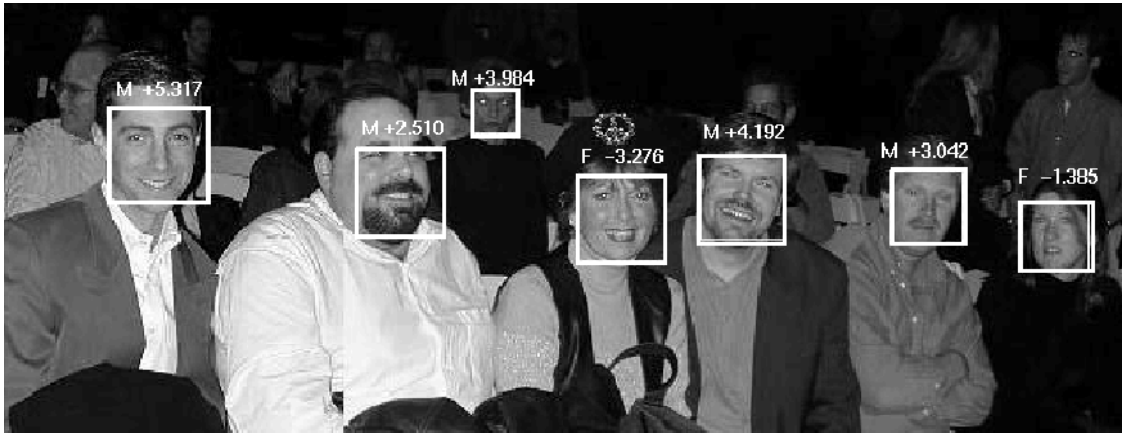
# Other detection tasks

Facial Feature Localization

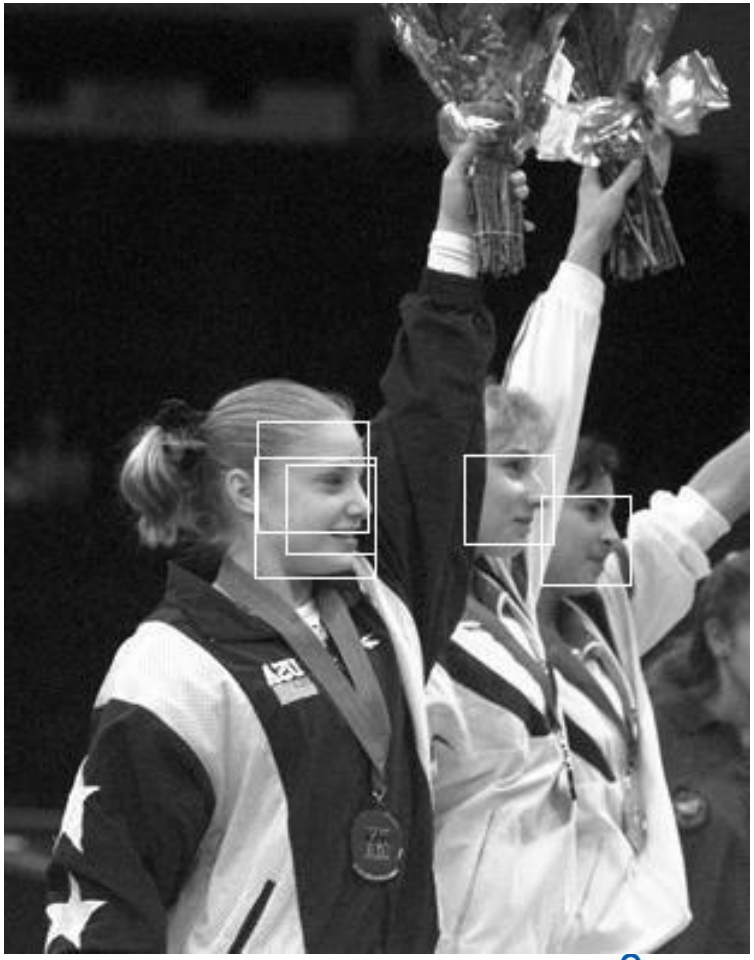


Profile Detection

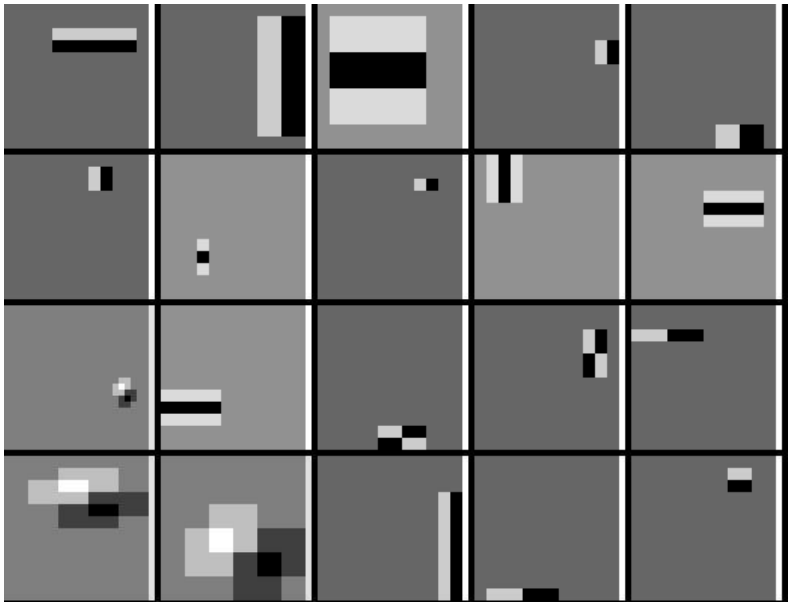
Male vs. female



# Profile Detection



# Profile Features



# Summary: Viola/Jones detector

- Rectangle features
- Integral images for fast computation
- Boosting for feature selection
- Attentional cascade for fast rejection of negative windows